

DATA MINING PROJECT

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Final Segmentation & Business Insights of ABCDEats Inc.

Group 03

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1. INTRODUCTION

In today's competitive landscape of food delivery services, understanding customer preferences is pivotal to success. ABCDEats. Inc, a dynamic food delivery platform, recognizes this challenge and embarks on a data-driven journey to enhance its customer segmentation strategy. By leveraging customer data collected over three months from three distinct cities, the company seeks to uncover meaningful patterns and insights that will enable personalized services and targeted marketing campaigns. The goal is to identify diverse customer segments by analyzing attributes such as purchasing behavior, demographic information, and culinary preferences. These insights will empower ABCDEats to tailor its offerings to better meet the needs and preferences of each customer group, ultimately fostering stronger customer engagement, satisfaction, and loyalty. This project involves a comprehensive exploration of the dataset, identification of key variables for segmentation, and the application of clustering techniques to delineate customer groups. Each segment will be thoroughly analyzed to highlight its unique characteristics, ensuring that the final segmentation strategy is not only data-informed but also aligned with ABCDEats' broader business objectives. By integrating Culinary-based, Behaviour-based, and Expenditure perspectives into the segmentation framework, ABCDEats aims to craft a holistic marketing strategy that resonates with its diverse customer base. This endeavor not only positions the company to optimize its operations but also paves the way for sustainable growth in the ever-evolving food delivery market.

2. DATA UNDERSTANDING

The dataset provided for ABCDEats Inc. encompasses a comprehensive array of customer data, capturing diverse behavioral, demographic, and temporal attributes over a three-month period. It includes 31,888 rows and 56 features, offering a good foundation for data-driven customer segmentation analysis. This dataset provides a wealth of insights into customer preferences and interactions. Key features include: Variables such as *customer_age* and *customer_region* provide foundational insights into the geographic and age-related distribution of the customer base. With an average customer age of 27.5 years (standard deviation of 7.16 years), the dataset reveals a predominantly young clientele, which could inform targeted marketing strategies. Metrics like *vendor_count* (average 3 vendors) and *product_count* (average 5.6 products per transaction) offer a glimpse into customer engagement with the platform. High variance in cuisine preferences, such as *CUI_American* and *CUI_Asian*, indicates diverse customer tastes, presenting opportunities for personalized recommendations. Features such as *DOW_0* to *DOW_6* (*Day of Week*) and *HR_0* to *HR_23* (*Hour of Day*) illuminate ordering patterns across time, highlighting peak engagement periods. These insights are crucial for optimizing operational and marketing efforts. Lastly, features like cuisine categories (*CUI_American*, *CUI_Italian*, etc.) enables the identification of high-value customers and preferences that may inform promotional strategies.

Further Insight: During our exploration of the dataset, we observed several intriguing patterns and trends that shed light on customer behaviors and preferences such as: On average, customers place their first order about 28.5 days into the dataset timeline. Interestingly, there is an average gap of 63.7 days between their first and most recent orders, suggesting diverse patterns of engagement over time.

As we developed deeper, certain days and hours stood out as peak ordering times. These patterns provide valuable insights into when customers are most active, helping identify opportunities for

targeted promotions and operational optimizations. We also noticed significant variability in spending across different cuisine categories. For example, while some categories like *CUI_American* and *CUI_Asian* were popular, their high variance suggests that customer preferences in these areas can be polarizing—people either love them or steer clear. These insights offer a glimpse into the rich potential of this dataset shown in fig.1 below. By digging deeper into these patterns, we aim to uncover actionable strategies that can enhance customer engagement and support data-driven decision-making for ABCDEats Inc.

	customer_age	vendor_count	product_count	is_chain	first_order	last_order	CUI_American	CUI_Asian	CUI_Beverages	CUI_Cafe
count	31161.000000	31888.000000	31888.000000	31888.000000	31782.000000	31888.000000	31888.000000	31888.000000	31888.000000	31888.000000
mean	27.506499	3.102609	5.668245	2.818866	28.478604	63.675521	4.880438	9.960451	2.300633	0.801163
std	7.160898	2.771587	6.957287	3.977529	24.109086	23.226123	11.654018	23.564351	8.479734	6.427132
min	15.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	23.000000	1.000000	2.000000	1.000000	7.000000	49.000000	0.000000	0.000000	0.000000	0.000000
50%	26.000000	2.000000	3.000000	2.000000	22.000000	70.000000	0.000000	0.000000	0.000000	0.000000
75%	31.000000	4.000000	7.000000	3.000000	45.000000	83.000000	5.660000	11.830000	0.000000	0.000000
max	80.000000	41.000000	269.000000	83.000000	90.000000	90.000000	280.210000	896.710000	229.220000	326.100000

Fig.1

3. DATA TREATMENT

1. Missing values: During our analysis, we identified missing values in several key columns of the dataset. To quantify this, we calculated both the total missing values and their corresponding percentages, as shown in Annex 1. For columns with missing data, we carefully selected imputation methods based on their distribution and characteristics. For instance, *customer_age* and *first_order* were imputed using their median values, as their distributions, illustrated in Annex 2, indicated a left-skewed pattern without significant outliers. This approach ensured that the dataset retained its integrity and statistical reliability, allowing us to move forward with our analyses. ([Annex 1 & 2](#))

2. Outliers: In this stage, our main goal was to clean the dataset so that it provides a solid foundation for clustering analysis. We used four key techniques: **visual inspection, manual thresholding, IQR-based outlier removal, and a combined manual-IQR filtering approach** : At first, we plotted histograms for key features such as *vendor_count*, *product_count*, *Orders on Weekend*, *Orders in the morning*, and some others, so that we could identify potential outliers that significantly deviate from the majority of the data. We observed that most of the features exhibited a right-skewed distribution with long tails, indicating the presence of outliers. This step helps us to provide a baseline understanding of the data and informs the choice of outlier removal methods([Annex 3](#)). Then we implemented **manual thresholds** based on domain knowledge and visual observations to filter out extreme or unrealistic values. Specifically, *vendor_count* values above 20, *total_amount_spent* above 1000, and *Orders on Weekend* exceeding 30 were capped as they represent unrealistic or extreme scenarios. To remove unreasonably high values, it helps us to ensure the dataset remains aligned with practical business expectations. This approach successfully keeps 97.68% of the rows after removing outliers. We also applied the **IQR rule** to systematically detect and exclude outliers by defining bounds based on the 25th (Q1) and 75th (Q3) percentiles. And we noticed that this approach was problematic since it keeps only 30.32% of rows after removing outliers. We then **combined filtering techniques**, and after applying them, 97.8% of the data is retained, which is the same as manual thresholds. As a result, we finally used manual thresholding. After that we again used the manual threshold technique for those features that are still not working for the outlier remover. At first, using

.value_counts(), we identified how many rows remain without outliers. This step helped us make informed decisions about appropriate thresholds for balancing outlier removal and data retention. Based on these observations, we defined a dictionary of thresholds to cap extreme values, and using the .clip() method, thresholds were replaced with the maximum allowable value without removing entire rows.

4. FEATURE ENGINEERING

In our efforts to extract deeper insights from the customer data, we introduced several new features to better understand their behaviors and preferences. To analyze ordering patterns, we created variables such as *Orders in the morning*, *Orders in the afternoon*, and *Orders in the night*, which capture the number of orders during specific time periods. For example, morning covers 7:00-13:00, afternoon spans 13:00-20:00, and night includes 20:00-7:00. Additionally, we calculated the total number of orders placed on weekends and weekdays through the features *Orders on the weekend* and *Orders on the week being weekdays Monday to friday*.

To understand spending habits, we grouped cuisine-related data into broader categories. For instance, Asian-related spending includes cuisines like Indian, Chinese, Japanese, and Thai, while snacks and beverages combine expenditures on items like café products, desserts, and street snacks. Similarly, total spending on American, European (Italian), and other cuisines was calculated and aggregated into a single feature, *total_amount_spent*. We also introduced metrics, such as *order_frequency*, which measures how often customers place orders by calculating the difference between their first and last orders divided by the total number of orders. To segment customers based on age, we created an *age_group* feature that categorizes customers as Young (0-24 years), Middle-aged (25-50 years), or Senior (51-100 years). In addition, we calculated several ratios to analyze the proportion of spending across different cuisines compared to the total spending which was made to every group of cuisine mentioned before. For example, *Asian_ratio* represents the percentage of total spending on Asian cuisines. Lastly, we created a feature, *avg_vendor_use*, by dividing the number of vendors by the total product count to understand how many products customers typically purchase from each vendor. In these new variables, we found some missing values in ratios more specifically 138 which we removed because the dataset would still retain a lot of information.

4.1 Analysis Before Clustering Techniques:

Before proceeding with clustering, we conducted a comprehensive exploratory analysis to identify patterns and relationships within the data. This involved examining customer behavior, spending trends, and feature correlations to uncover actionable insights. Our analysis of ordering behavior revealed that 71.2% of orders were placed on weekdays, while only 28.8% occurred on weekends (Annex 5). Considering there are more weekdays than weekend days, weekday ordering still dominates, likely driven by work-related or routine activities. Further, we segmented orders based on the time of day and found that the afternoon (13:00-20:00) accounted for the majority of orders (47.7%), followed by the morning (7:00-13:00) at 33.9%, and the night (20:00-7:00) at 18.4%. (Annex 6) These findings are pivotal for optimizing operational and marketing strategies to target high-demand periods effectively. In addition, to understand customer spending preferences, we grouped cuisine-related features into broader categories such as Asian, American, European,

Snacks & Beverages, and Other cuisines(Annex 7). Asian cuisine emerged as the most popular, attracting the highest number of customers, followed by American cuisine and Snacks & Beverages. Spending analysis across payment methods highlighted that card payments dominated both total and average spending, while digital payments showed lower overall engagement.(Annex 8)

We also categorized customers into age groups (*Young, Middle-Aged, and Senior*) and observed that middle-aged customers placed the most orders, indicating their significant contribution to the platform's activity (annex 9). Additionally, we examined the usage of promotions across payment methods, noting distinct preferences for discounts and freebies among different customer segments(Annex 10).Lastly, we leveraged correlation matrices to explore relationships among key features, including ordering patterns, cuisine spending, and engagement metrics. Strong correlations were identified between weekday orders and total orders, while weaker relationships between certain cuisines and ordering patterns highlighted diverse customer preferences(Annex 11). These correlations informed the selection of attributes for clustering and ensured that significant variables were included in the segmentation analysis.Finally,this detailed analysis provided valuable insights into customer behavior, spending habits, and preferences, forming a good foundation for applying clustering techniques and deriving meaningful customer segments.

5. DATA SCALING

We scaled the numerical features in our dataset to ensure uniformity across all variables before applying clustering techniques. This step was essential to prevent features with larger numerical ranges from dominating the clustering process. Using the StandardScaler from the Sklearn library, we transformed all numerical columns to have a mean of 0 and a standard deviation of 1. The process began by identifying numerical features, excluding identifiers like *customer_id*, and including variables such as *vendor_count*, *product_count*, *Orders on Weekend*, *Orders in the morning*, and *various cuisine ratios*. These features were selected because they represent diverse aspects of customer behavior and engagement.After scaling, we observed that all features were standardized effectively. The summary statistics confirmed this, with means close to 0 and standard deviations of approximately 1 across all scaled features. For instance, features like *Asian_ratio* and *avg_vendor_use* were scaled to ensure they contributed equally during the clustering process. Additionally, categorical variables such as *last_promo* and *payment_method* were encoded and scaled appropriately to integrate seamlessly into the dataset.After all, this transformation ensured that our dataset was balanced and prepared for clustering, allowing the analysis to identify meaningful patterns and customer segments without bias from feature magnitudes.

6. CLUSTERING

6.1 Segmentation Analysis

Before applying clustering techniques, we conducted a series of preparatory steps to ensure our dataset was well-structured and the features were meaningful for segmentation. First, we categorized the dataset's features into four main groups to analyze and understand their relationships:

1. **Behavioral Features:** These variables, such as *Orders on Weekend*, *Orders on week*, *Orders in the morning*, *Orders in the afternoon*, *Orders in night*, *order_frequency*, and *avg_vendor_use*,

help us identify how customers engage with the service and their habits. They provide insights into peak ordering times, preferences for weekdays or weekends, and general ordering behavior.

2. **Food Preference Features:** This group includes features like *American Cuisine*, *Asian Cuisine*, *European Cuisine*, *Snacks & Beverages*, and *Other Cuisine*. These variables enable us to understand customer preferences for various cuisines, offering insights into their food choices and potential cultural or regional influences.
3. **Expenditure Features:** Variables such as *total_amount_spent*, *American_ratio*, *Asian_ratio*, *European_ratio*, *Snacks_Beverages_ratio*, and *Other_Cuisine_ratio* provide a detailed view of how customers allocate their spending across different categories. These features help in identifying high-value customers and spending patterns.
4. **Vendor Features:** Features like *vendor_count*, *product_count*, and *is_chain* reflect the diversity and nature of customer orders. They provide insights into whether customers prefer chain restaurants, how many products they order, and the variety of vendors they interact with.

After categorizing the dataset's features, we calculated and visualized correlation matrices to explore the relationships among features within each feature group. For **behavioral Features**, we observed moderate correlations between weekday orders and other time-based variables like afternoon orders and order frequency. *avg_vendor_use*, however, showed weaker correlations, highlighting its distinct behavior. **Food Preference Features** exhibited relatively low correlations, indicating that customers have varied preferences for different cuisine types, with little overlap in spending patterns. Furthermore, for **expenditure Features**, stronger correlations were noted between some spending ratios, such as *Asian_ratio* and *Other_Cuisine_ratio*, suggesting similarities in customer spending across these categories. High correlations between *vendor_count*, *product_count*, and *is_chain* revealed a connection between order diversity and the type of restaurant (chain or non-chain) for Order Features ([Annex 12](#)).

6.2 Clustering Analysis and Feature Optimisation:

Here, our main goal was to identify the optimal number of clusters for each segment and streamline the feature set by removing redundancies or low-contributing features, leading to an efficient clustering solution:

1. Behavioral Features Analysis: For the analysis, we first applied K-Means clustering on behavioral features such as *Orders on Weekend*, *Orders in the morning*, and *Average Vendor*. The table shows the cluster centroids for each feature across different clusters ([Annex 13](#)). These centroids represent the average values of the features for the respective clusters, giving us insights into the customer behavioral patterns within each group. After that, using the R^2 plot, we analyzed the clustering performance for different methods—**complete**, **average**, **single**, and **Ward's method**. The R^2 value indicates the proportion of variance explained by the clustering method, with Ward's method performing the best. Based on the elbow method from the R^2 plot, we selected 4 clusters as the optimal solution for further analysis ([Annex 14](#)). Then we analyzed the correlation matrix ([Annex 12](#)) to check the relationship between features. It confirmed that no highly correlated variables (*correlation above 0.8*), ensuring no redundancy. Among the features, "*Orders on Week*" had the highest feature importance, while "*Average Vendor Use*" was the least significant in influencing clustering ([Annex 16](#)). Silhouette Analysis: Silhouette scores for different cluster configurations (4, 5, and 6 clusters) were calculated to evaluate clustering quality. While we observed the highest silhouette score (0.2181) for 6 clusters, indicating better-defined group separation for this configuration ([Annex 15](#)) & ([Annex 15A](#)). Hierarchical Clustering: We further applied hierarchical clustering with 6 clusters using Ward's method and Euclidean distance. This clustering configuration was merged with the K-Means results to create a combined clustering output. The importance of

features was recalculated after clustering, where *"Orders in the Afternoon," "Orders in Night," and "Orders on Week"* showed the highest importance values. This step confirmed the relevance of the clustering approach and validated the chosen number of clusters as 6 (Annex 16) .

2. Food Preferences Features Analysis: Similar to the Behavioural Features, we applied the K-means algorithm to this feature set. The table illustrates the cluster centroids(Annex 17) represent the average preference values for each feature, providing insights into customer cuisine preferences. Using the R^2 plot, we compared clustering performance across methods like **complete**, **average**, **single**, and **Ward's method**. Among these, Ward's method performed the best, explaining the highest variance. Additionally, based on the elbow method, 4 clusters were selected for further analysis(Annex 18). We also applied a correlation matrix (Annex 12) with a threshold of 0.8, which revealed no strong correlations existed between the features, ensuring they provided unique information. Among the features, *"American Cuisine"* had the highest importance ($R^2 = 0.458633$), and *"Snacks & Beverages"* had the lowest contribution ($R^2 = 0.001054$) (Annex 19B). These findings suggest that features like "American Cuisine" and "Other Cuisine" are critical for differentiating clusters, while lower-contribution features can be considered for removal to improve our cluster quality. Silhouette scores: We calculated for cluster configurations of 3, 4, 5, and 6 clusters. The analysis revealed that the highest silhouette score (0.2748) was observed for 3 clusters (Annex 19), indicating the best separation between clusters for this configuration(Annex 19). Although the silhouette score for 4 clusters (0.2724) was slightly lower than that for larger clusters(Annex 19A), the R^2 plot indicated 4 clusters as the best choice. Considering this, and balancing silhouette scores with interpretability, we selected 4 clusters as the optimal solution.

Hierarchical Clustering: We applied hierarchical clustering using Ward's method and Euclidean distance with 4 clusters. The hierarchical clustering labels were merged with the K-Means clustering output to refine and validate the results. After hierarchical clustering, the importance of features was recalculated. We found that "American Cuisine" and "Other Cuisine" showed the highest importance values, reinforcing their relevance in segmentation. Features like "Snacks & Beverages" were found to have minimal impact, supporting potential feature reduction.(Annex 19B)

3. Expenditure/Spending Features Analysis: We conducted a detailed analysis of expenditure-related features using K-Means clustering. The clustering process revealed distinct spending patterns among customer groups, with cluster centroids showcasing average feature values for each group, providing meaningful insights into how different clusters allocate their spending. Then to evaluate clustering performance, we used the R^2 metric across various linkage methods, including complete, average, single, and Ward's method. The plot shows that Ward's method provided the highest R^2 values, suggesting it was the most effective method for capturing variance in the data(Annex 21). Additionally, based on the elbow method, 4 clusters were selected for further analysis. Furthermore, we evaluated the quality of clustering by calculating silhouette scores for different numbers of clusters (3, 4, 5, and 6) (Annex 21). The best **silhouette score(0.2414)**(Annex 21A) was achieved with 3 clusters, although other configurations provided acceptable separations. Although the **silhouette score(0.1854)** for 4 clusters was slightly lower than that for larger clusters, the R^2 plot indicated 4 clusters as the best choice. Considering this, and balancing silhouette scores with interpretability, we selected 4 clusters as the optimal solution. We also constructed a correlation matrix (Annex 24) to evaluate relationships between spending features. The analysis revealed no significant multicollinearity, ensuring that the features were distinct and not redundant. We observed minor

correlations, such as between *Asian_ratio* and *Other_Cuisine_ratio* but did not affect clustering outcomes significantly. Using R^2 metrics (Annex 23), we analyzed the contribution of each feature to clustering effectiveness. The feature *total_amount_spent* had the highest contribution to the clustering process, followed by *Other_Cuisine_ratio* and *European_ratio*. In contrast, features such as **American_ratio(0.048553)** and **Snacks_Beverages_ratio(0.046540)** had lower contributions to the clustering process (Annex 23). To validate and enhance K-Means clustering, hierarchical clustering was applied using Ward's linkage and Euclidean distance with 4 clusters. Results confirmed the clustering patterns observed earlier, emphasizing the importance of *total_amount_spent* and *Other_Cuisine_ratio* in defining group separations.

Clustering Performance Comparisons: For **Snacks & Beverages**, we identified as having a low R^2 contribution (Annex 23). Removing it significantly improved **silhouette scores(0.181923)** which enhanced clustering quality and group distinctions. Additionally, correlation analysis showed no strong dependencies between **Snacks & Beverages** and other features. The feature was considered redundant due to its low importance in defining cluster boundaries. We noticed after removing **Snacks & Beverages**, the dataset became simpler and easier to interpret, particularly by emphasising the roles of more important key features like *Asian Cuisine* and *American Cuisine*.

4. Vendor/Order Features Analysis: The vendor-related features analyzed include *vendor_count*, *product_count*, and *is_chain*. A similar methodology as in previous analyses was applied, with K-Means clustering used to group these features into clusters. The centroids of these clusters represent the average characteristics of each group, helping us understand vendor patterns. Then, the R^2 metric was evaluated for different clustering methods (complete, average, single, and ward). Here, K-means initialisation and hierarchical clustering validated Ward's method with 4 clusters as optimal. Then, the highest silhouette score of 0.2242 (Annex 22) & (Annex 22C) reinforced this decision. Furthermore, the correlation matrix (Annex 12) showed no high correlations (above 0.8) among the vendor features, ensuring no feature redundancy. We found that *is_chain* had the highest importance, contributing significantly to the variance explained by clustering (Annex 12) (Annex 22D) and other features like *product_count* and *vendor_count* also showed meaningful contributions. (Annex 22D)

In conclusion, We applied KMeans clustering because it was shown the best comparison of other algorithms, and we determined the optimal number of clusters using a combination of R^2 , silhouette analysis, and correlation evaluations. The analysis revealed 6 optimal clusters for the behavioral feature set, while 4 clusters were optimal for the food preferences, expenditure, and vendor-related feature sets.

6.3 Contingency Table and Insights

We created a contingency table (Annex 25) to measure how well different clustering approaches aligned and whether clusters influenced specific combinations. To achieve this, we calculated the means for features such as order frequency, spending ratios, and vendor engagement. This step allowed us to capture the average behavior of customers in each cluster. Additionally, we calculated the mean of the metric features for each combination of cluster labels (*behavior*, *cuisine*, *spending*, and *vendor clusters*), and we captured the center or representative point of each cluster group. These centroids summarize the average characteristics of the grouped samples. We applied Hierarchical clustering to organize the centroids into a tree-like structure, allowing us to understand how

clusters are related at different levels. This was particularly useful for merging related clusters or exploring sub-cluster structures. Using KMeans clustering, we segmented the dataset into behavioral, cuisine, spending, and vendor-related clusters, enabling us to uncover distinct patterns. For example, Cluster 2 demonstrated significantly higher mean values for *"Orders on Weekend"* (2.11), *"Orders on Week"* (2.51), and *"Orders in the Morning"* (2.65), indicating a group of highly active customers(Annex 25). In contrast, Cluster 0 showed the lowest mean values across all ordering metrics, including *"Orders on Week"* (-0.66) and *"Order Frequency"* (-1.28), representing customers with infrequent or minimal engagement. Similarly, Cuisine clusters revealed strong distinctions in preferences for cuisine types. For instance, Cluster 2 had a high mean value for *"American Cuisine"* (1.98) and *"Asian Cuisine"* (1.36), suggesting a strong affinity for these cuisines, whereas Cluster 1 showed a significantly high mean for *"Other Cuisine"* (1.08)(Annex 25). Furthermore, spending clusters showed differences in spending habits. Cluster 1 spent more, especially on *"Asian Cuisine"*, while Cluster 3 spent the least, showing a cost-conscious approach. Vendor clusters revealed interaction levels, with Cluster 2 engaging the most with vendors (mean *"vendor_count"*: 3.22), while Cluster 0 had low interaction.

The contingency table (Annex 25) also showed overlaps between some clusters, like Clusters 1 and 3, which shared similar spending and behavior patterns. Other clusters, like Cluster 4, were unique and required separate approaches. Overall, this analysis provided simple and clear insights into customer groups, helping to create more targeted strategies for marketing and operations.

6.4 Final Analysis for Clusters:

The clustering analysis reveals insightful segmentation of customer behavior, spending patterns, cuisine preferences, and vendor interactions. These clusters enable us to design personalized strategies and optimize resource allocation to enhance customer satisfaction and maximize business value. For **Behaviour Perspective**, the "Behavior Perspective" analysis highlights distinct customer behavior patterns across clusters, providing actionable insights for business strategies. Cluster 4 represents the largest and most valuable segment, characterized by very high engagement across all ordering patterns, strong preferences for European cuisine, snacks, and beverages, and significant total spending and order frequency. Similarly, Cluster 3 is a key target group with high engagement in vendor and product interactions, above-average orders across all timeframes, strong preferences for American and Asian cuisines, and substantial spending. Together, these clusters should be prioritized for loyalty programs and promotional strategies. Conversely, Clusters 1 and 2 exhibit minimal engagement across vendors, products, ordering patterns, and cuisines, representing low-value or infrequent customers who may benefit from targeted reactivation campaigns or deprioritization based on profitability(Annex 26). Cluster 0 reflects balanced, average customer behavior with moderate engagement and spending, while Cluster 5 represents stable but unremarkable behavior patterns. This segmentation enables businesses to tailor strategies that maximize value from high-contributing groups while optimizing efforts for lower-engagement segments(Annex 26) .

For **Food Perspective**, the "Food Preferences Perspective" analysis revealed distinct customer segments based on their food preferences and ordering behaviors. Cluster 2 and Cluster 3 emerge as high-value segments, with Cluster 2 representing adventurous food lovers who prefer diverse cuisines, and Cluster 3 focusing intensely on snacks and beverages with exceptionally high spending and order frequency. **These clusters are ideal for tailored premium offerings, loyalty programs, and promotions.** Cluster 1, a moderately valuable segment, highlights consistent ordering patterns and preferences for snacks and European cuisine, suggesting opportunities to expand offerings in these areas(Annex 27). Conversely, Cluster 0, the largest group, consists of low-engagement customers with

minimal spending and weak preferences across all food categories, indicating the need for cost optimization or targeted re-engagement strategies. This segmentation underscores the influence of food preferences on customer behavior and provides actionable insights for targeted marketing and operational improvements.

For **Expenditure/spending Perspective**, Cluster 0 represents customers with lower activity, characterized by fewer orders, lower order frequency, and reduced spending. These customers exhibit a lower preference for "Other Cuisines" compared to other clusters. Cluster 1, in contrast, shows moderate values across most metrics, with customers placing a reasonable number of orders and spending a moderate amount. They exhibit a balanced preference across different cuisines. Cluster 2 includes customers with a high number of orders, order frequently, and spend significantly more. They show a strong preference for "Other Cuisines" and explore a variety of options. Cluster 3, similar to Cluster 2, this cluster shows high activity and spending. However, they have a stronger preference for "*European Cuisine*" compared to other clusters. The clusters appear to capture different customer segments based on their spending behavior, order frequency, and cuisine preferences. The analysis suggests that there are segments with high activity and spending, segments with specific cuisine preferences, and segments with lower activity and spending ([Annex 29](#)).

For **Order/Vendor Perspective**, Cluster 0, customers exhibit the highest levels of vendor interactions and a strong preference for snacks and beverages. Additionally, they place many orders on weekends. We observed from the bar chart, cluster 0 is the largest group, which represents a majority of customers who are highly active. Then, Cluster 1 includes customers with moderate activity. These individuals show balanced ordering behavior across different times of the day and prefer American cuisine. They tend to interact less with vendors compared to Cluster 0, but they still show consistent engagement. Cluster 2 represents less active customers. They spend less, order less often, and interact with vendors the least. This cluster includes cost-conscious or low-engagement customers. Cluster 3 includes highly active customers. They spend more and prefer Asian and European cuisines. They also order more frequently, especially in the mornings and at night ([Annex 28](#)).

For **Combined Perspective**, Cluster 0 is the largest cluster with the highest frequency. Customers in this group exhibit balanced behaviour with moderate values across most features, including vendor count, orders, and cuisine preferences. These customers exhibit consistent ordering behaviour. Cluster 1 has fewer members but shows higher engagement with vendors. These customers have a stronger preference for snacks, beverages, and specific cuisines, such as American and Asian cuisines. Cluster 2 represents customers with low activity. They place fewer orders, spend less, and interact minimally with vendors. So, this cluster includes cost-conscious or inactive customers. Cluster 3 includes customers with higher activity during specific times, such as weekends and mornings. They spend more on Asian and European cuisines compared to other clusters. Cluster 4 is a distinct group with high activity in ordering snacks & beverages. These customers represent a smaller, distinct group. Cluster 5 is the smallest cluster. Customers in this group have specific patterns in spending and ordering, focusing on particular times and cuisine preferences ([Annex 30](#)).

Business insights for clusters

Cluster Number	Cluster Name	Description	Recommended Actions
Cluster 0	Balanced Customers	Customers with moderate spending patterns and a preference for traditional cuisines. Their ordering behavior is steady, with slight emphasis on weekend orders.	Focus on consistent engagement through stable offerings and traditional menu options.
Cluster 1	High-Value Explorers	High-value customers with diverse preferences and frequent orders. They exhibit the highest spending and a willingness to explore premium options.	Offer loyalty programs, upselling opportunities, and exclusive premium offers.
Cluster 2	Conservative Routine-Oriented	Routine-driven customers who prefer European cuisines and have lower spending habits. They are consistent in weekday orders, preferring budget-friendly and reliable meal options.	Maintain affordability and reliability to sustain engagement.
Cluster 3	Consistent Spenders	Moderate spending across a mix of cuisines. This cluster is dependable, with no extreme behaviors or preferences.	Cater to their consistency with standard offerings and periodic promotional campaigns.
Cluster 4	Bargain Hunters	Responsive to discounts and affordable offerings, such as snacks and beverages. Their spending is closely tied to promotional activities.	Regularly offer promotional campaigns, discounts, and low-cost bundles to retain and boost activity.

7. CONCLUSION

In conclusion, our analysis has provided a comprehensive understanding of customer behavior, preferences, and spending patterns across various clusters. By utilizing clustering techniques, we have identified distinct groups and their unique characteristics, enabling the creation of targeted strategies. Each cluster presents opportunities to personalize offerings, optimize resources, and enhance operational efficiency. Implementing the outlined recommendations will allow the business to deliver tailored experiences that cater to specific customer needs, increasing satisfaction and loyalty. Moreover, operational strategies aligned with cluster-specific behaviors will improve resource utilization and ensure smooth functioning. The integration of advanced technological solutions will not only enhance customer convenience but also provide actionable insights for sustained business growth. Together, these strategies pave the way for building stronger customer relationships and achieving long-term success.

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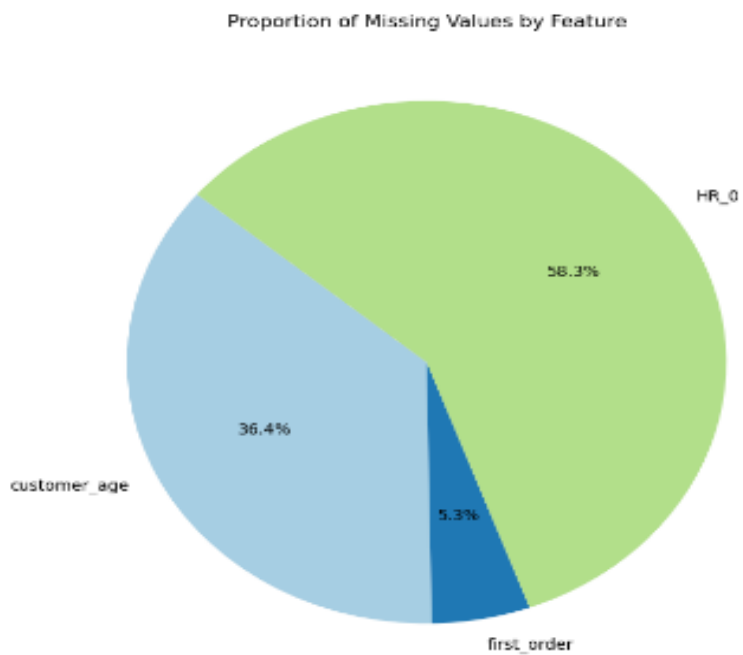
- Scikit-learn Documentation. (n.d.). Retrieved from <https://scikit-learn.org/stable/documentation.html>
- Towards Data Science. (n.d.). Articles on clustering techniques and best practices. Retrieved from <https://towardsdatascience.com>
- Analytics Vidhya. (n.d.). Comprehensive guides on feature engineering and segmentation analysis. Retrieved from <https://www.analyticsvidhya.com>
- Medium: Machine Learning & AI Blogs. (n.d.). Topics covering clustering methodologies and segmentation strategies. Retrieved from <https://medium.com/topic/machine-learning>
- Kaggle Data Science Tutorials. (n.d.). Resources on clustering and feature engineering. Retrieved from <https://www.kaggle.com/learn>

4. Authentic AI Research Pages

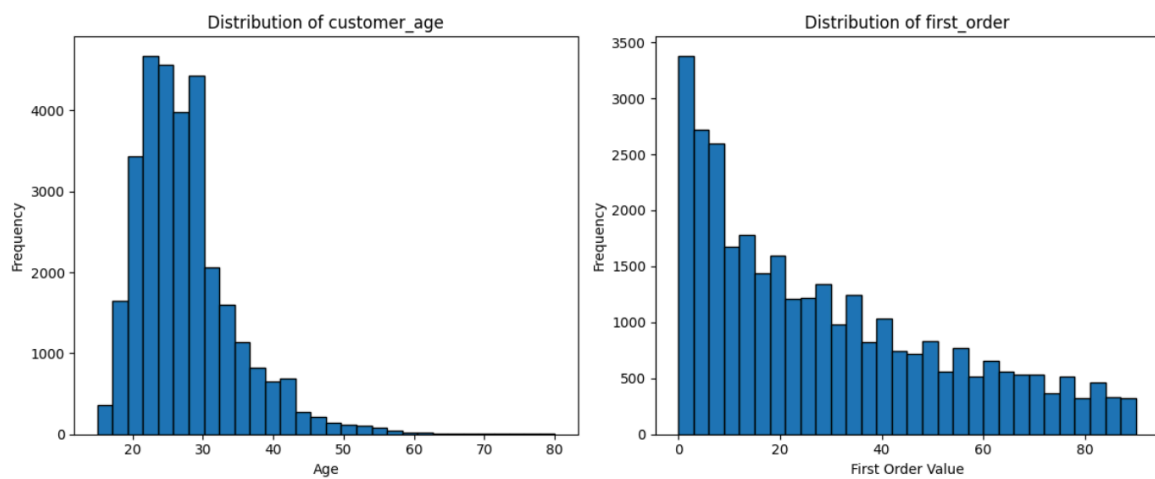
- Google AI Blog: Insights into AI and clustering algorithms. Retrieved from <https://ai.googleblog.com>
- OpenAI Blog: Understanding machine learning clustering applications. Retrieved from <https://openai.com/research>
- IBM Developer: AI and Data Clustering Techniques. Retrieved from <https://developer.ibm.com>

Appendix

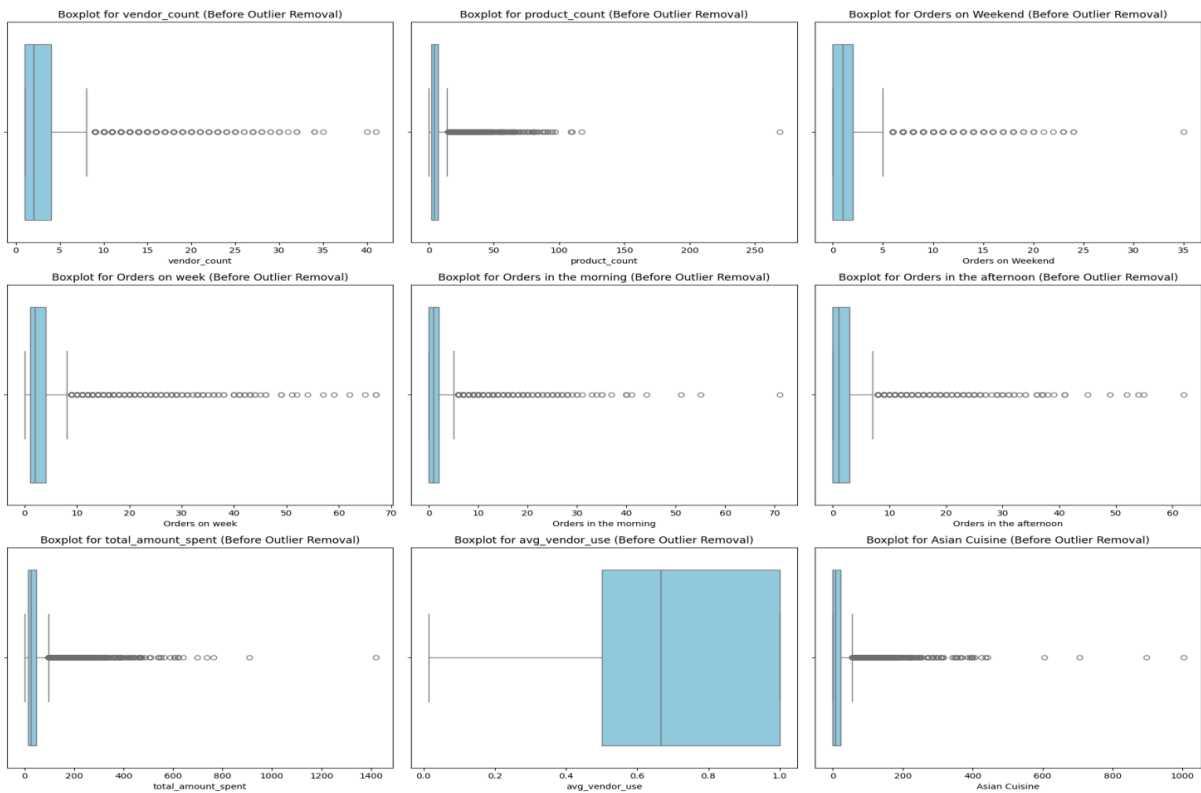
Annex 1 & 2 -Distribution of missing values



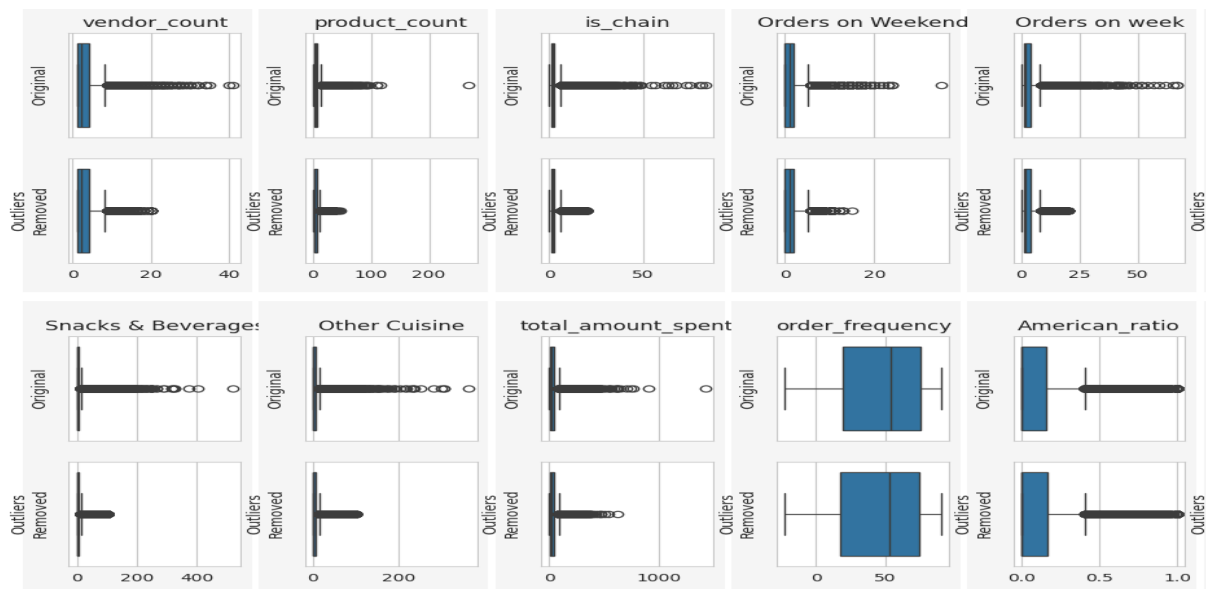
Annex 2 – Distributions of missing values



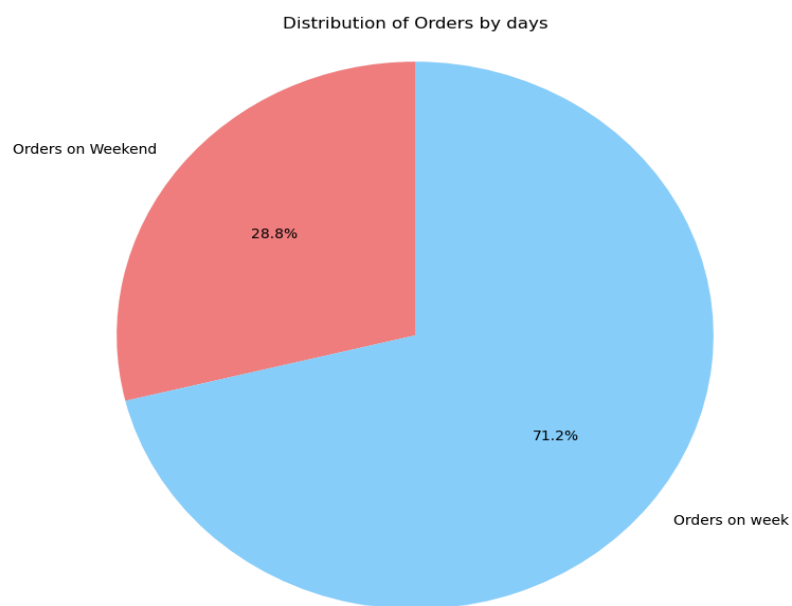
Annex 3 – Features before Outliers removal



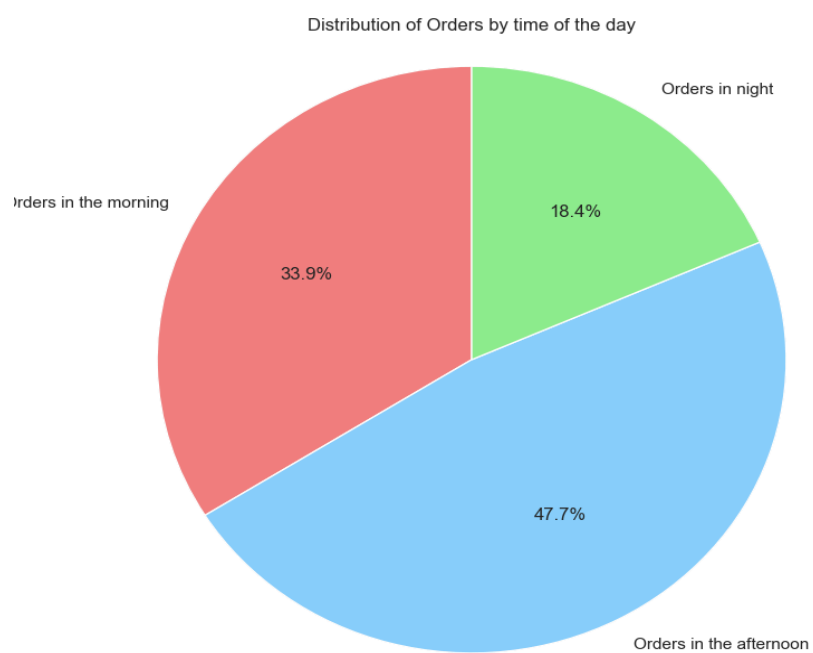
Annex 4 – Features after Outliers removal



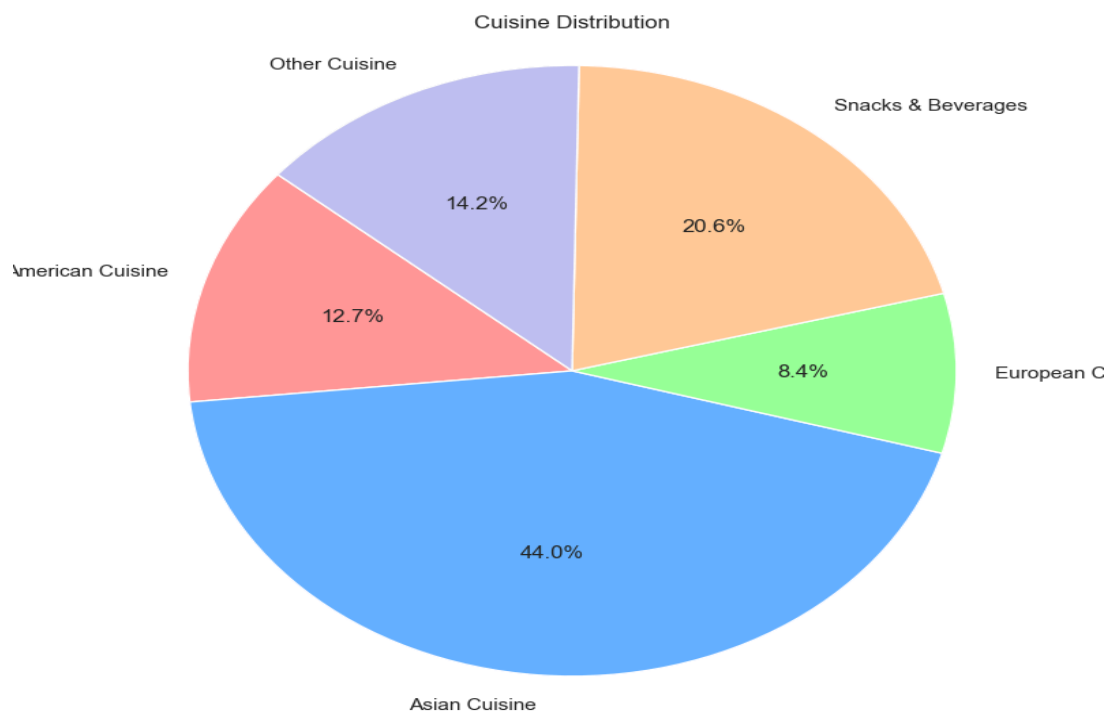
Annex 5- Distribution of orders by day



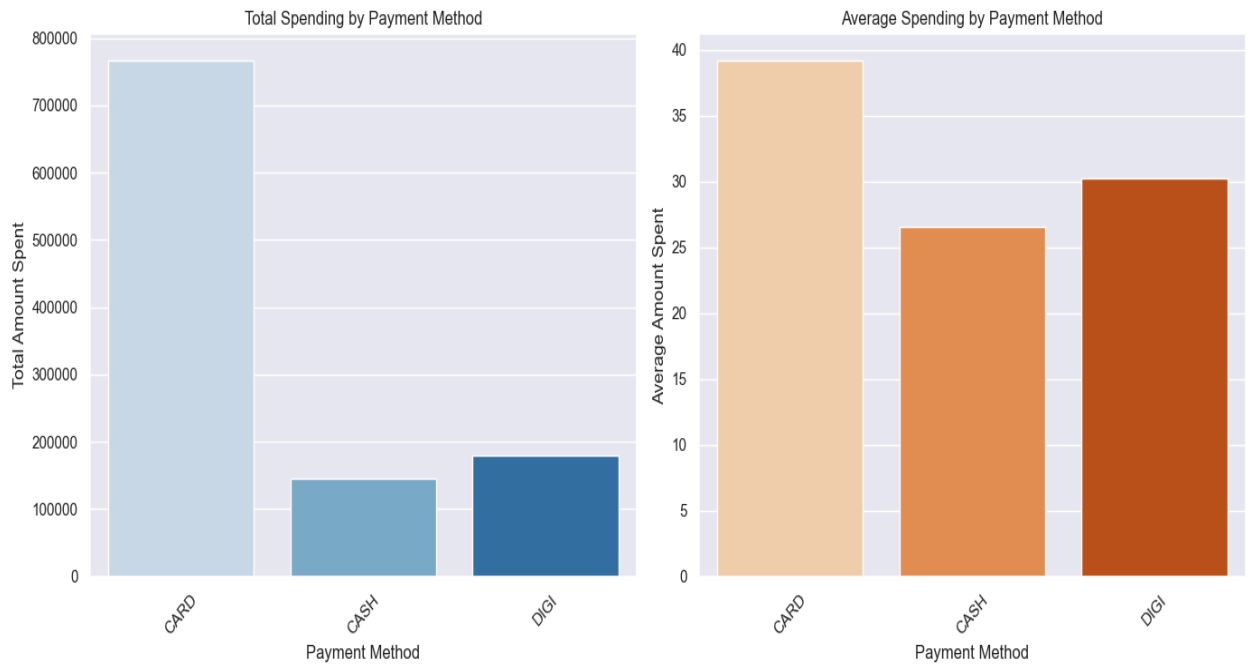
Annex 6- distribution by time of the day



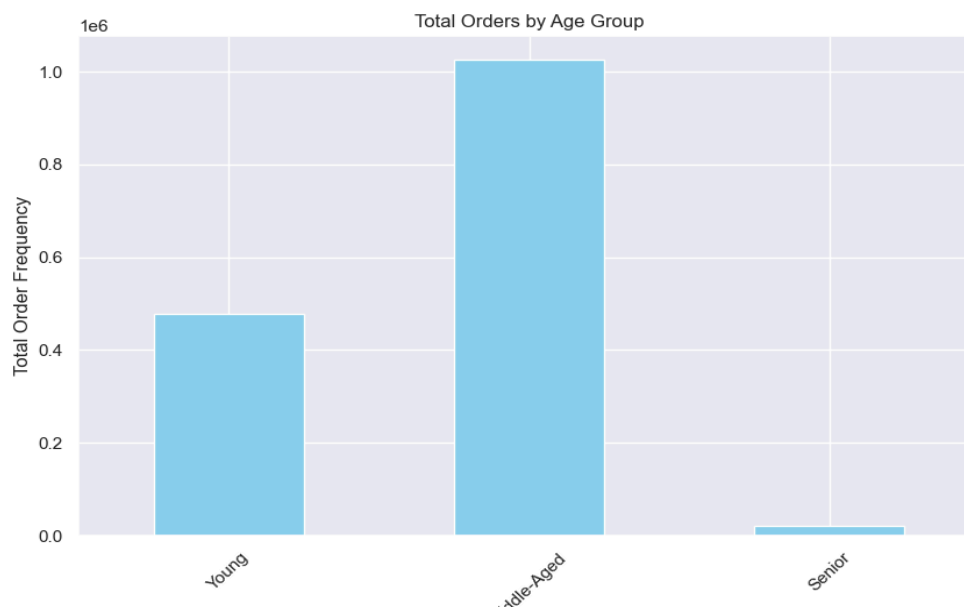
Annex 7- distribution by cuisine



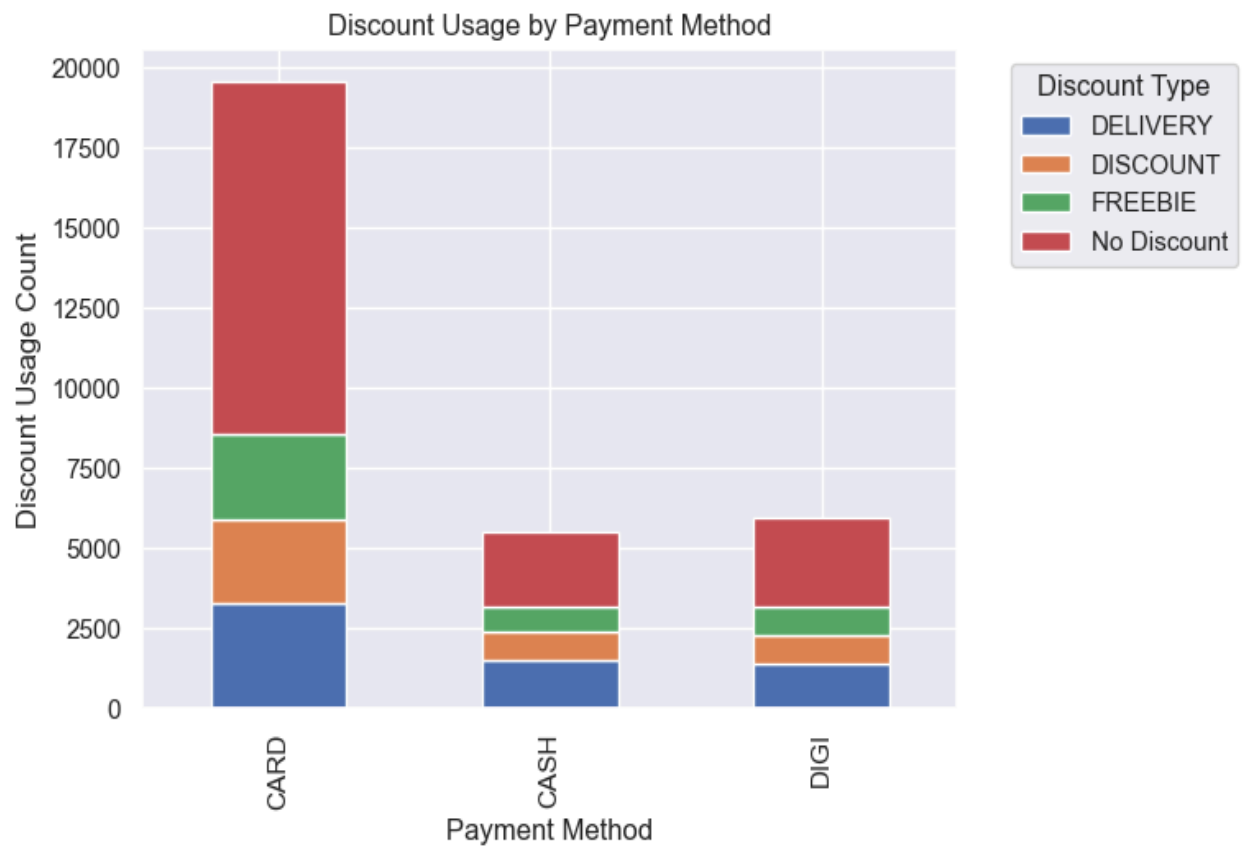
Annex 8- Amount spent by Payment method



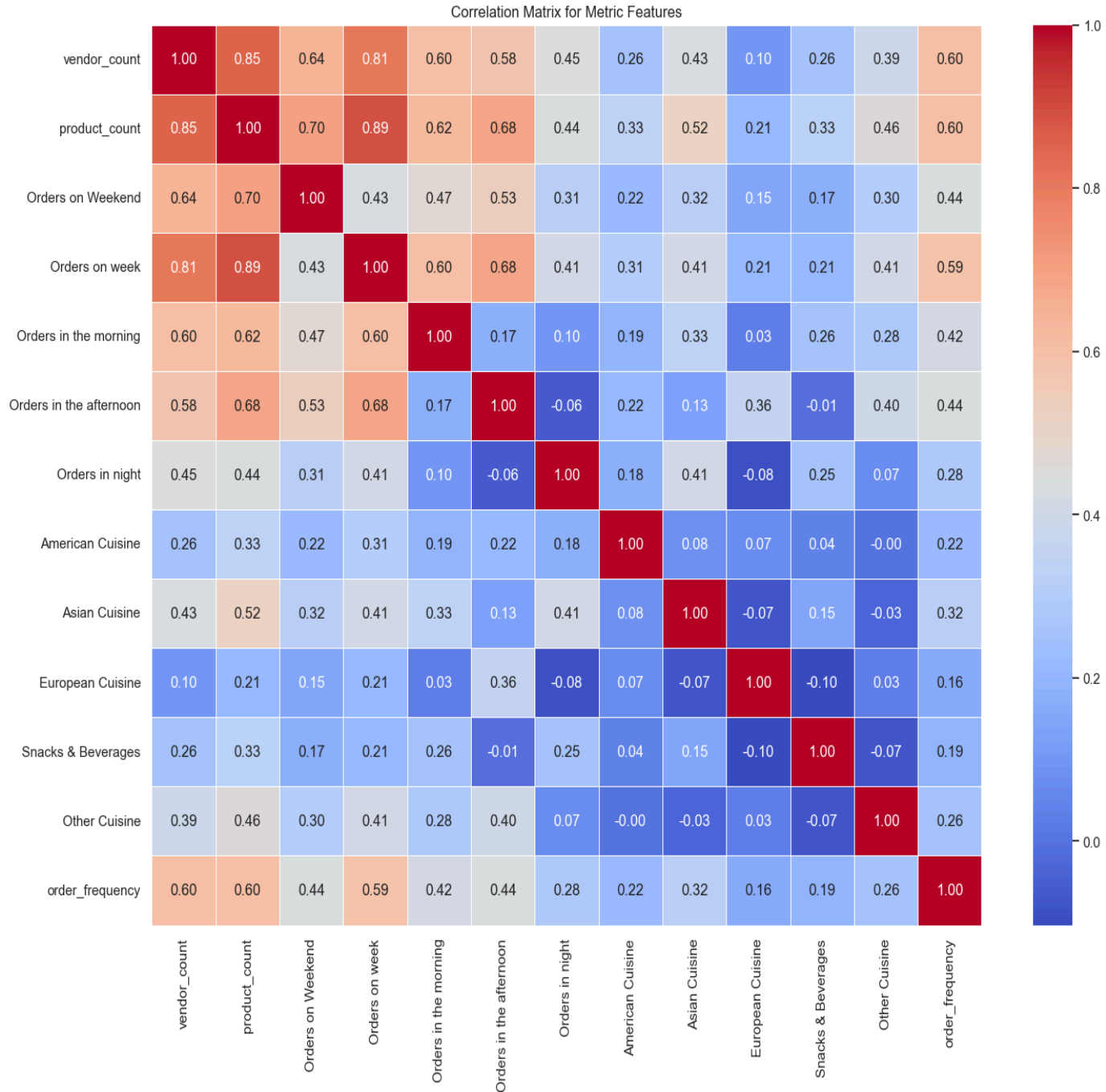
Annex 9 - Distribution by age group



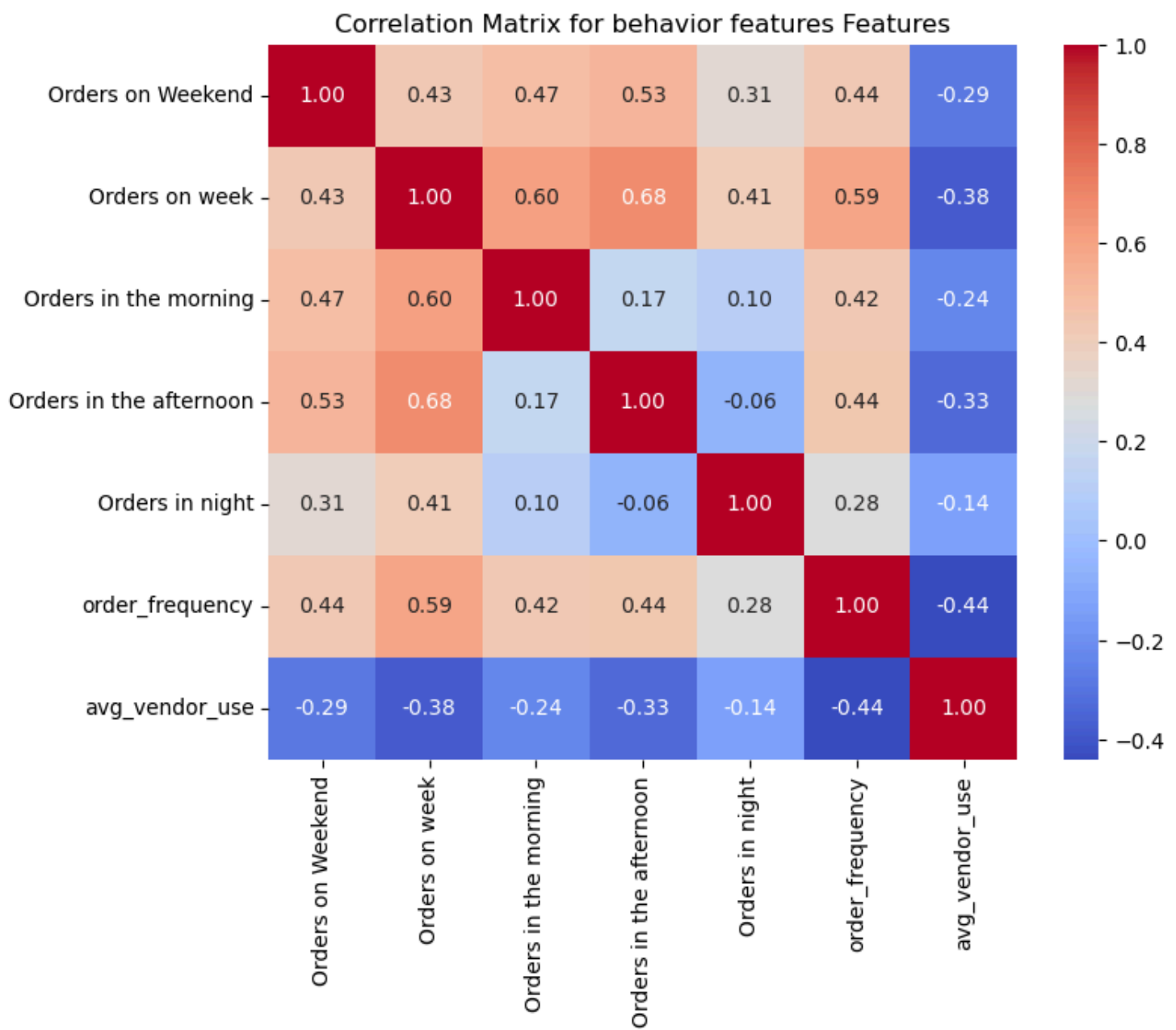
Annex 10 - Discount Usage by Payment Method

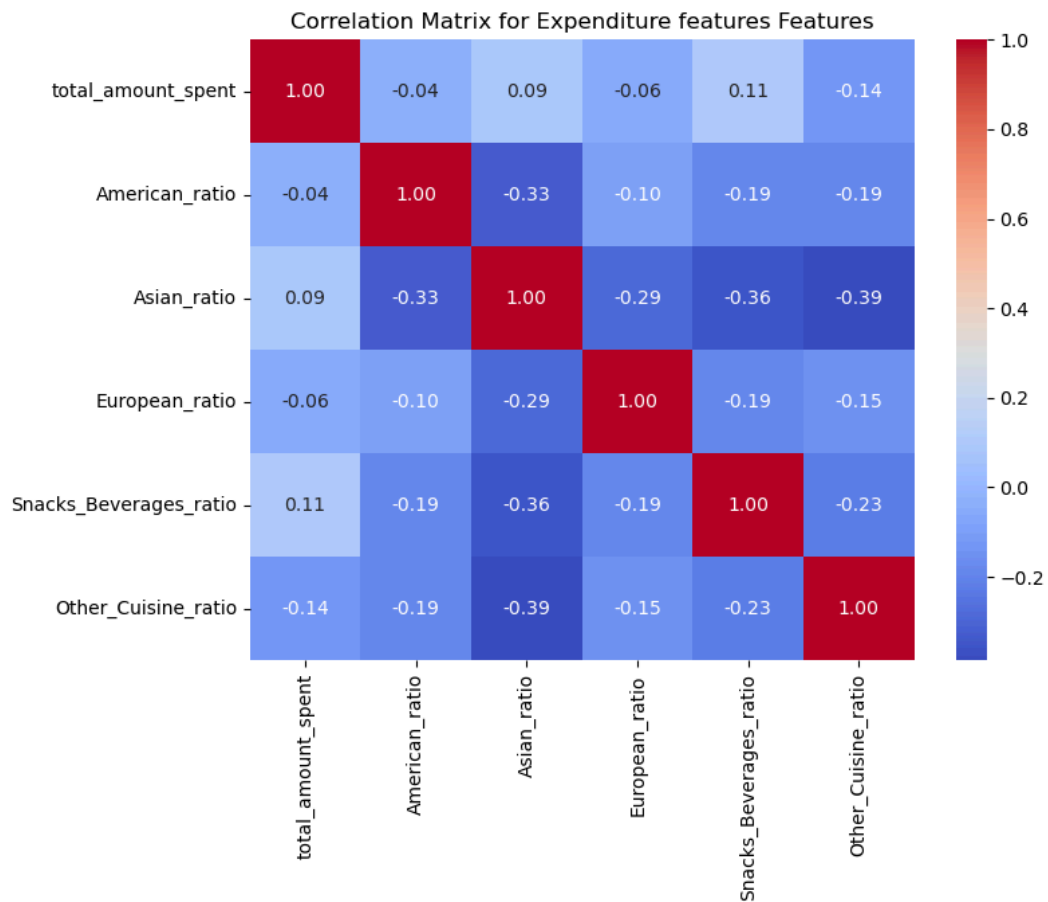


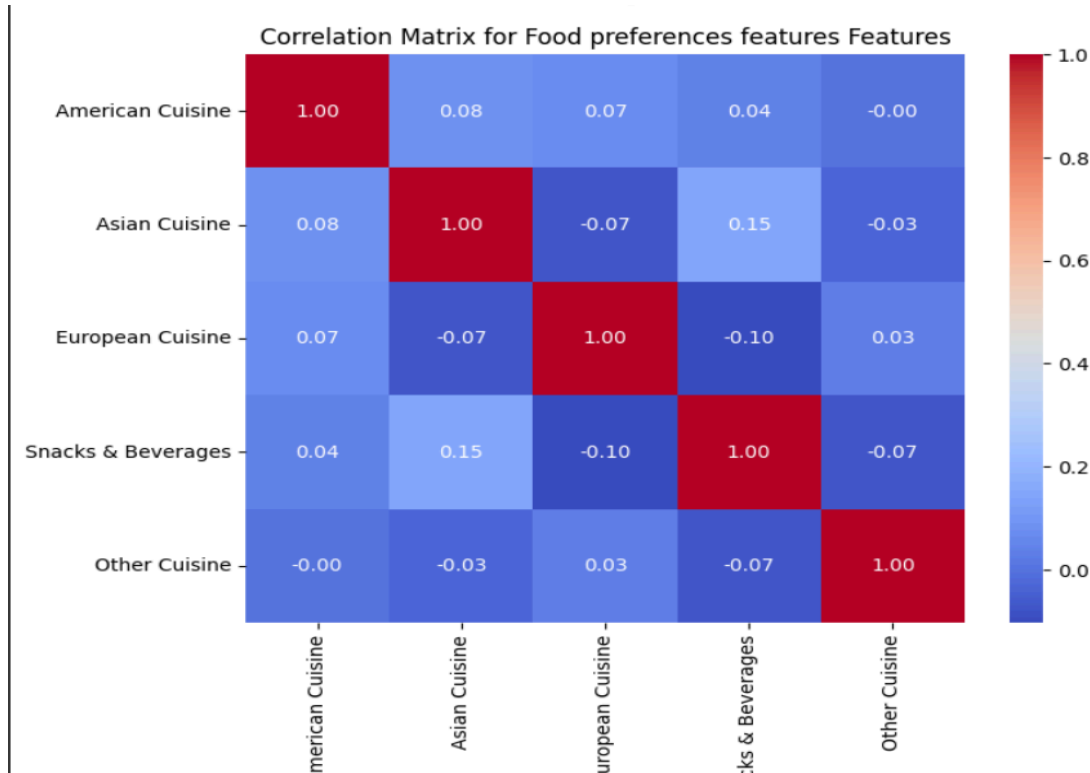
Annex 11- Correlation matrix in metric features



Annex 12- Correlation matrices for each segment





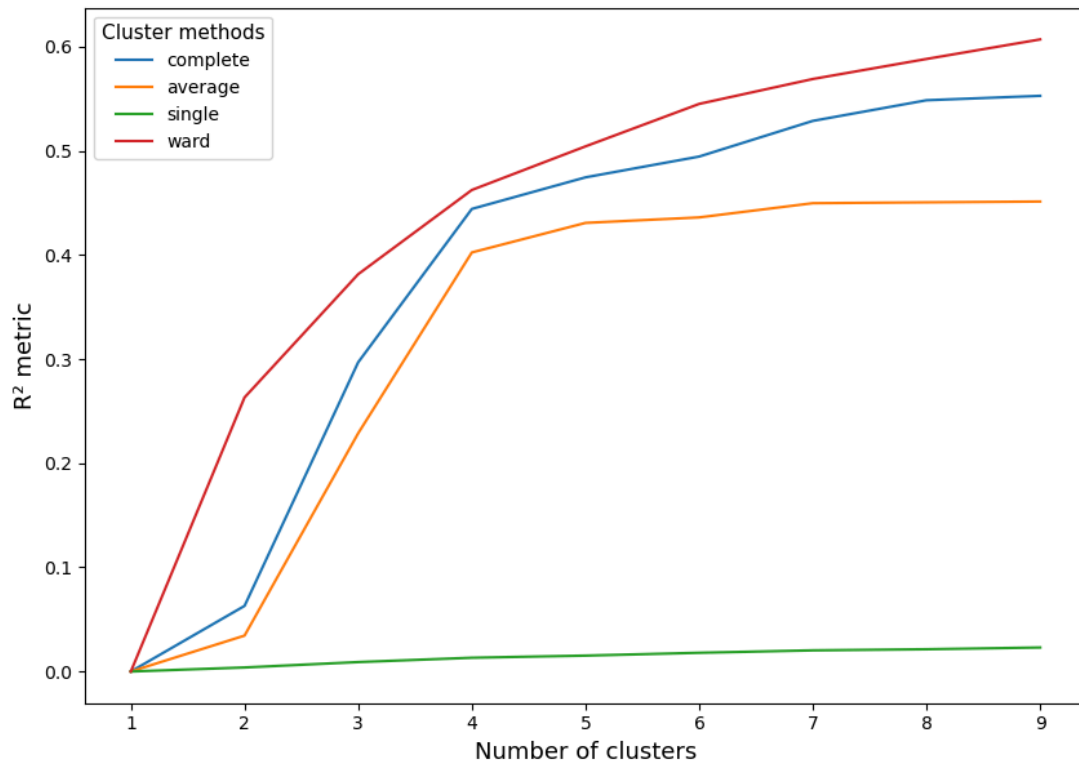


Annex 13- Distance of variables from each centroid

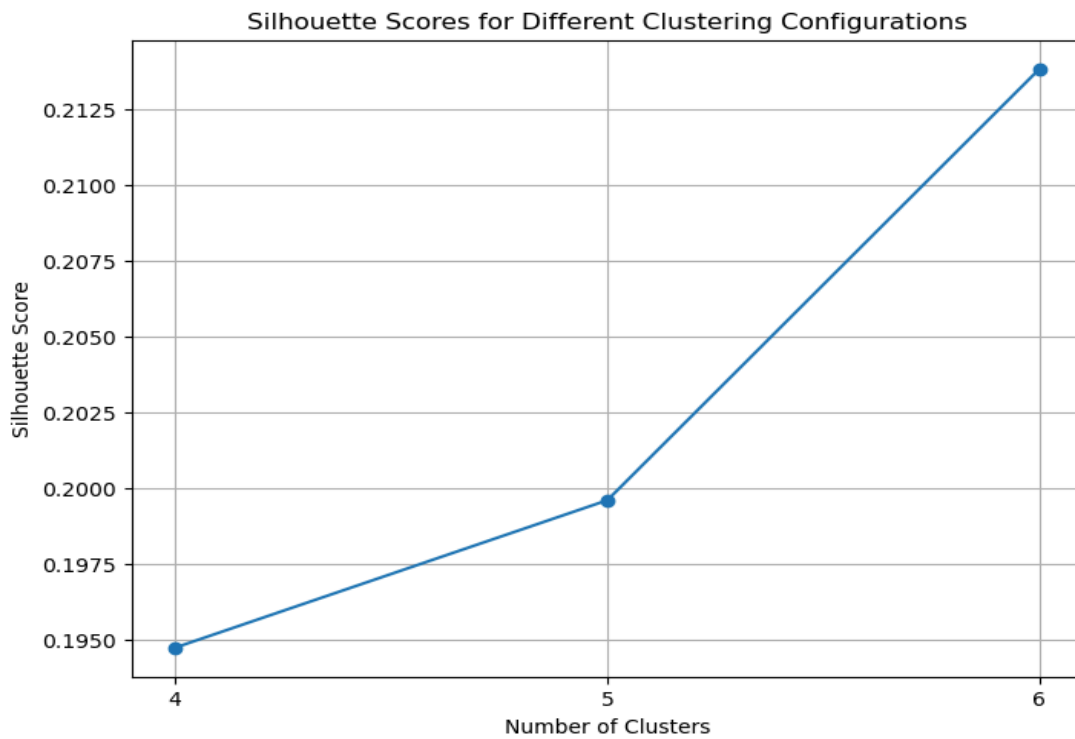
	Orders on Weekend	Orders on week	Orders in the morning	Orders in the afternoon	Orders in night	order_frequency	avg_vendor_use
CONTROLN							
0	-0.094029	-0.290780	-0.148124	0.066005	-0.543077	-0.882784	-0.004926
1	0.620842	0.222332	0.433980	0.650355	-0.543077	0.866203	-2.004782
2	1.335713	1.450691	2.180293	0.632648	0.414229	1.153724	-0.310470
3	-0.808900	0.935358	-0.148124	-0.723246	2.328842	0.649500	-0.821278
4	-0.808900	-0.663952	-0.730228	-0.358976	-0.543077	-1.660612	1.173853
...	...	...	...	...	...	...	...
1995	-0.808900	1.243374	1.339476	-0.595077	0.892882	1.200936	-0.144218
1996	0.620842	3.316553	4.896779	0.915969	-0.543077	1.295360	-1.486459
1997	-0.094029	0.455565	2.180293	-0.783958	-0.543077	0.339789	-0.482571
1998	0.825091	0.348944	-0.647070	0.855257	0.892882	1.188526	0.542587
1999	2.050584	6.053151	3.926606	5.590767	-0.543077	1.329353	-1.969558

2000 rows × 7 columns

Annex 14- R2 for behaviour variables in each linkage method
R² plot for various clustering methods



Annex 15- Silhouette scores by clusters in behavioral features



Annex 16 - importance of features in behavior features

```

Orders on week          0.591848
Orders in the afternoon 0.451552
Orders in night         0.329955
Orders in the morning   0.281894
Orders on Weekend       0.243423
order_frequency         0.160479
avg_vendor_use          0.090173
dtype: float64

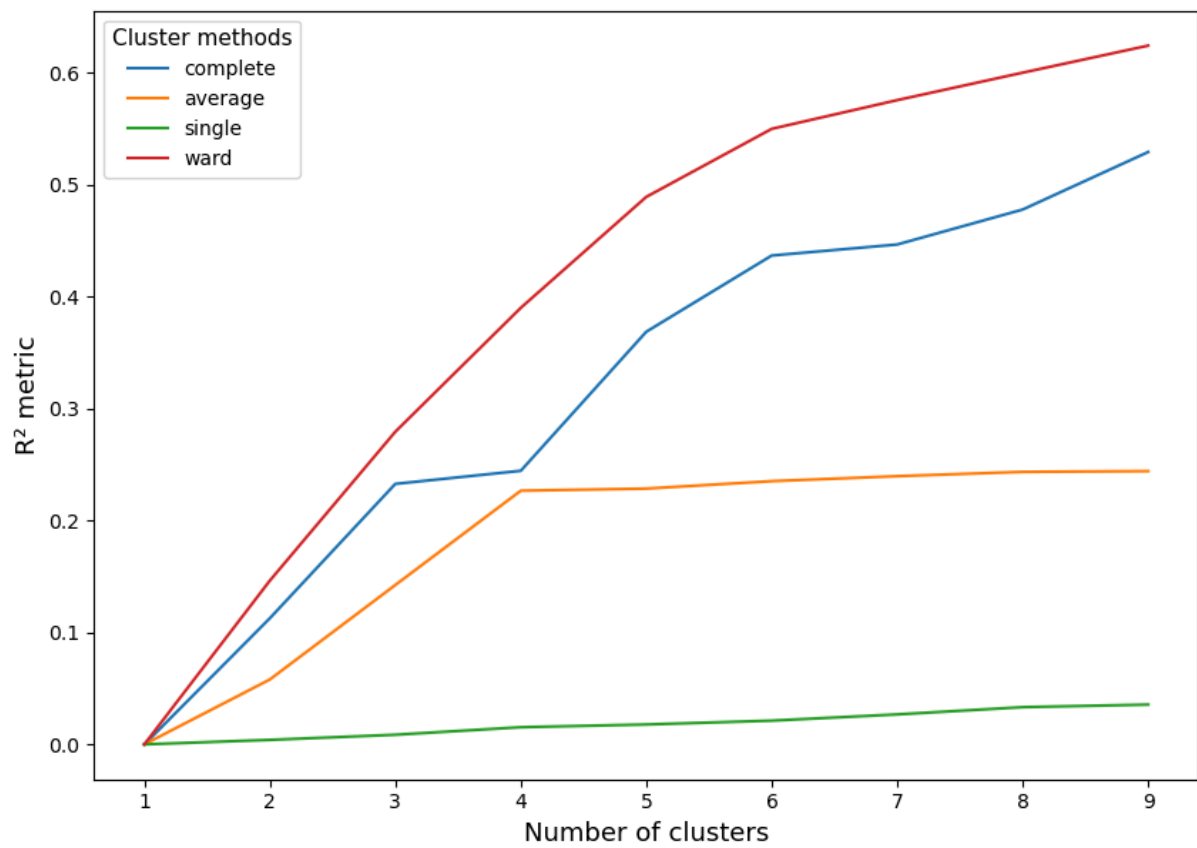
```

Annex 17- distance from centroids to each variable in food preference

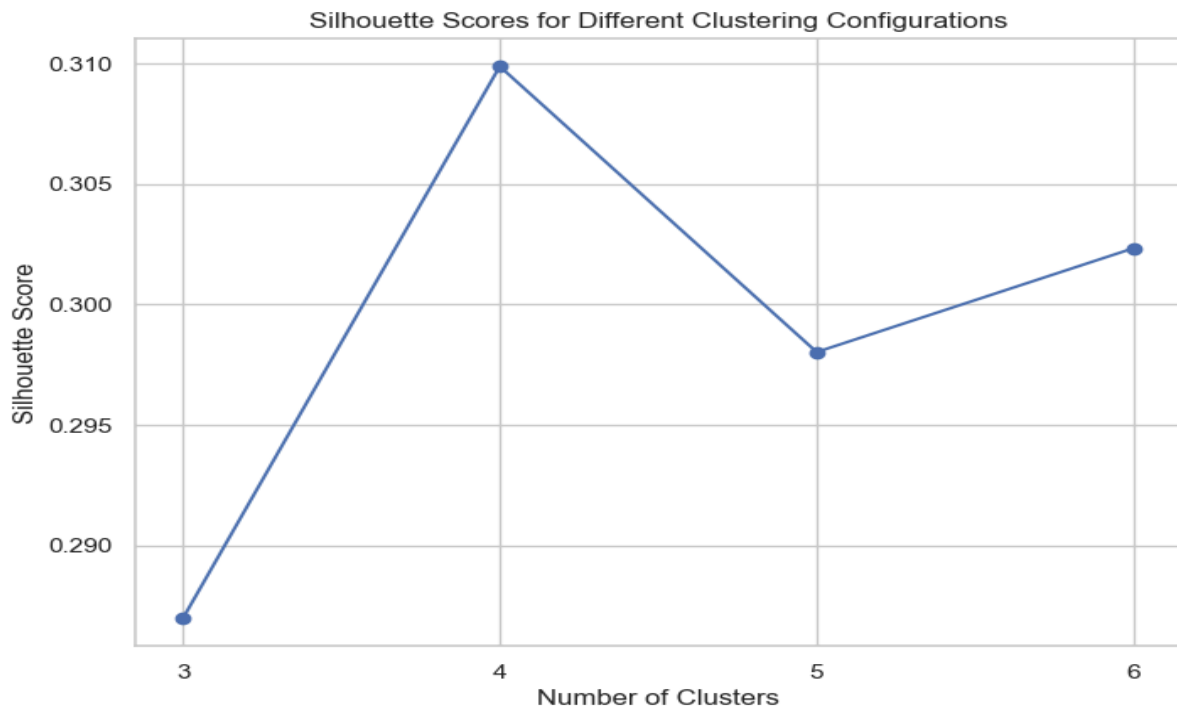
	American Cuisine	Asian Cuisine	European Cuisine	Snacks & Beverages	Other Cuisine
CONTROLN					
0	0.070590	-0.624293	-0.350768	-0.438510	-0.178136
1	-0.443473	1.890287	-0.352038	-0.440340	-0.453418
2	-0.443473	-0.150631	-0.352038	2.633427	-0.455182
3	0.202643	-0.487428	4.002349	-0.404638	-0.455182
4	3.026517	-0.148350	1.154286	-0.441718	3.876738
...	...	...	...	...	...
1995	1.275286	-0.113239	0.457643	-0.187840	-0.042803
1996	-0.443473	0.388788	3.935688	0.513544	2.993897
1997	-0.443473	-0.567629	2.607643	0.177370	-0.455182
1998	-0.103185	-0.599976	-0.352038	-0.089311	0.648964
1999	0.737099	7.355971	-0.352038	2.966294	-0.455182

Annex 18 - R² for food features in each linkage method

R² plot for various clustering methods



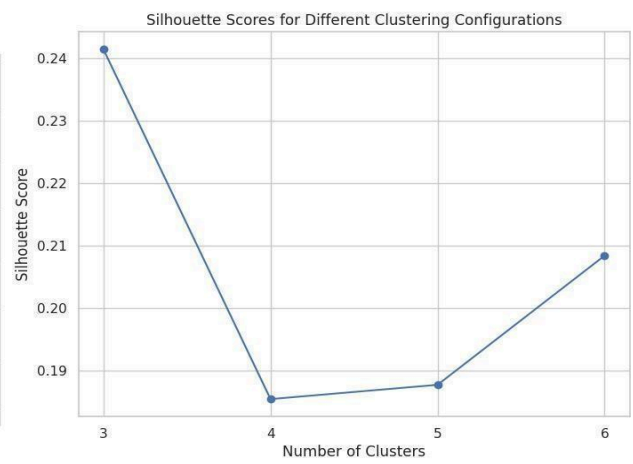
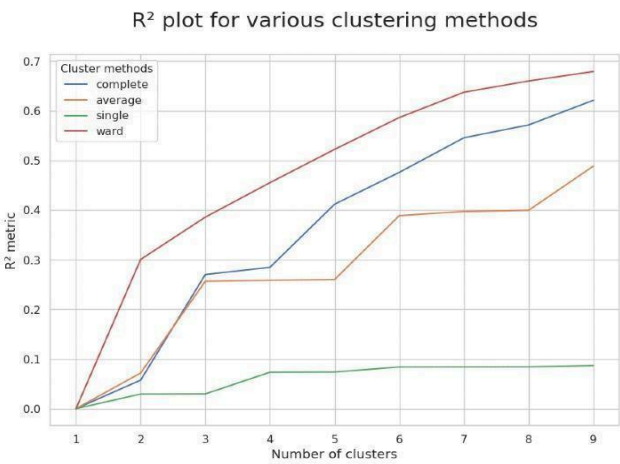
Annex 19- Silhouette scores by clusters in food preference



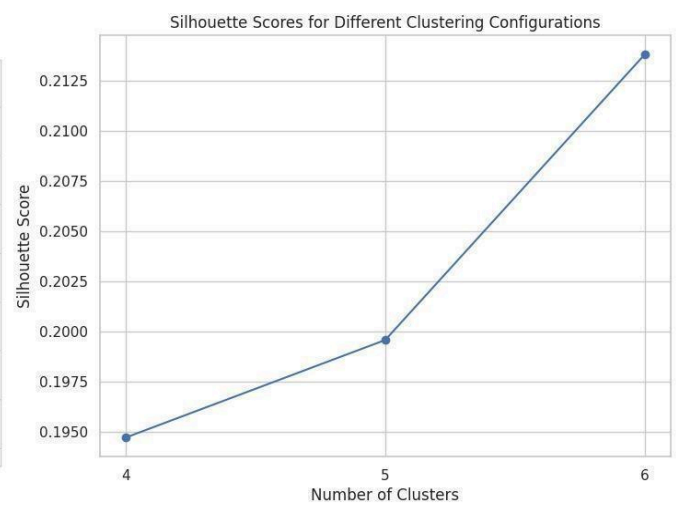
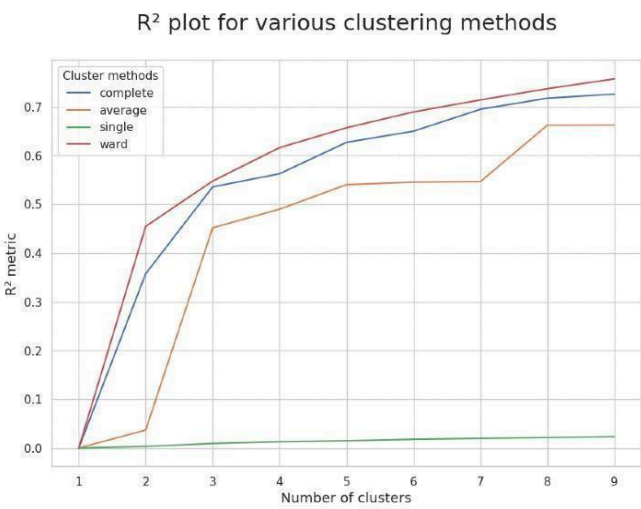
Annex 20 - importance of features in food preference features

Variable	threshold
Vendor_count	20
Product_count	50
Orders_on_weekend	30
Total_amount_spent	1000
Avg_vendor_use	0,1
Asian_cuisine	500
Orders on week	20
Orders in the morning	10
Orders in the afternoon	15

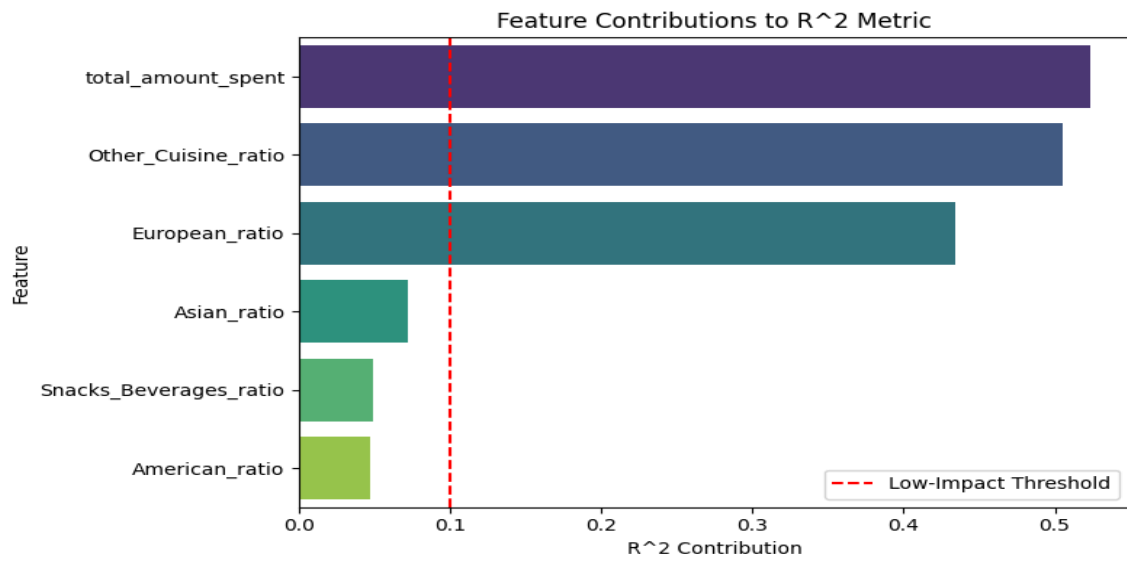
Annex 21 – Spending Features Analysis



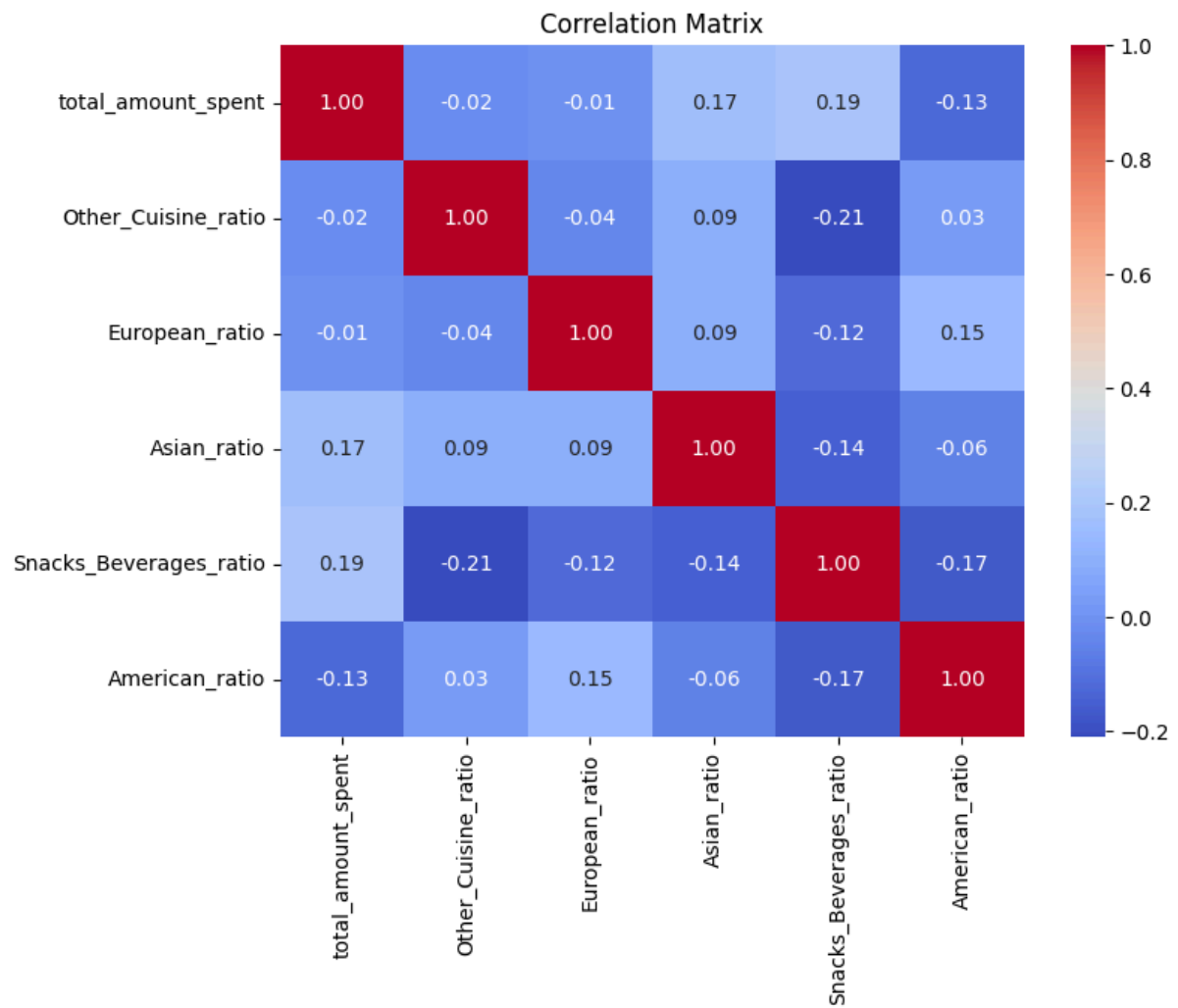
Annex 22 - Vendor Feature Analysis



Annex 23 - Spending Features Contributions to R-square



Annex 24 - Correlation matrix of spending features



Annex 25 - Contingency tables

```
df_cluster.groupby('behaviorCluster')[behavior_features].mean()
```

	Orders on Weekend	Orders on week	Orders in the morning	Orders in the afternoon	Orders in night	order_frequency	avg_vendor_use
behaviorCluster							
0	-0.512223	-0.657125	-0.491215	-0.487736	-0.321710	-1.275444	1.162275
1	0.605342	0.766279	-0.027609	1.451859	-0.312989	0.878020	-0.809802
2	2.106454	2.508587	2.645976	1.983784	0.284477	1.075820	-0.917058
3	-0.400199	-0.462994	-0.404621	-0.315801	-0.245683	-0.328048	-0.822916
4	-0.057014	-0.083935	0.225982	-0.260825	-0.047419	0.600543	0.352646
5	0.889695	1.178789	0.298261	-0.215776	2.932881	0.917155	-0.568930

```
df_cluster.groupby('CuisineCluster')[Food_preferences_features].mean()
```

	American Cuisine	Asian Cuisine	European Cuisine	Other Cuisine
CuisineCluster				
0	-0.226604	-0.127437	0.008227	-0.365972
1	-0.251800	-0.309099	-0.170428	1.082996
2	1.975257	1.355984	0.080434	-0.239319
3	0.101895	0.107671	0.459339	4.333158

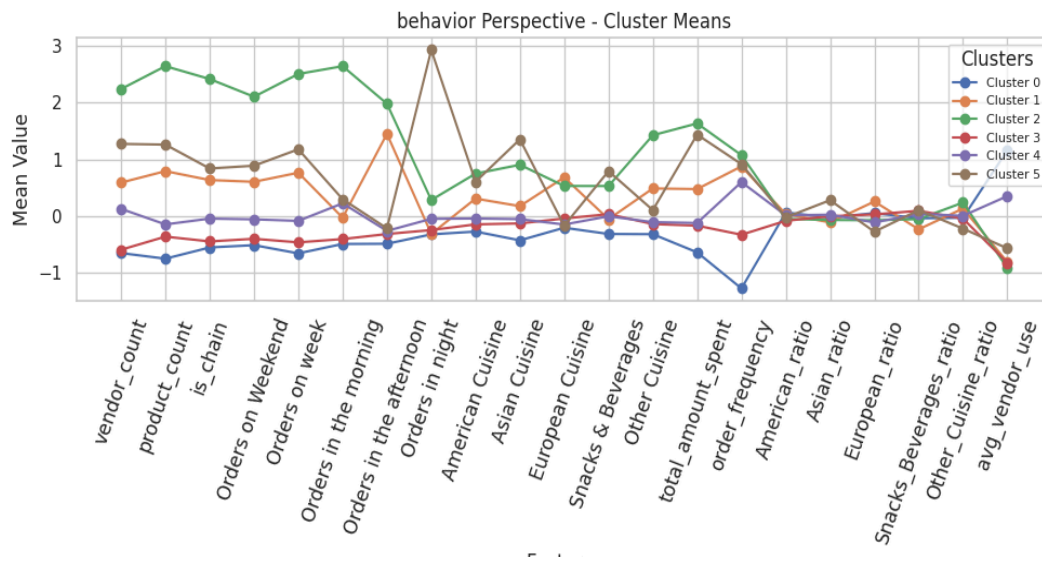
```
df_cluster.groupby('SpendingCluster')[Expenditure_features].mean()
```

	total_amount_spent	American_ratio	Asian_ratio	European_ratio	Snacks_Beverages_ratio	Other_Cuisine_ratio
SpendingCluster						
0	0.249648	-0.334728	-0.692638	-0.350141	1.920819	-0.464430
1	-0.257115	0.696154	-0.751905	-0.259844	-0.426855	0.979095
2	0.130396	-0.314491	1.074858	-0.299912	-0.412758	-0.435122
3	-0.199053	-0.297841	-0.840955	2.871383	-0.517326	-0.462841

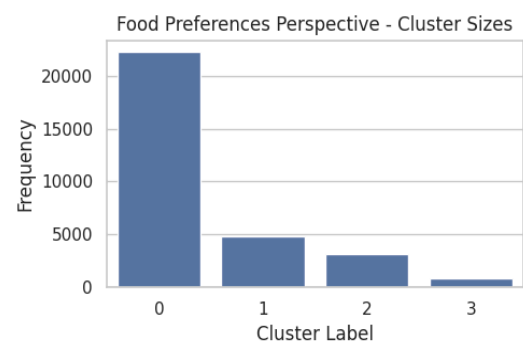
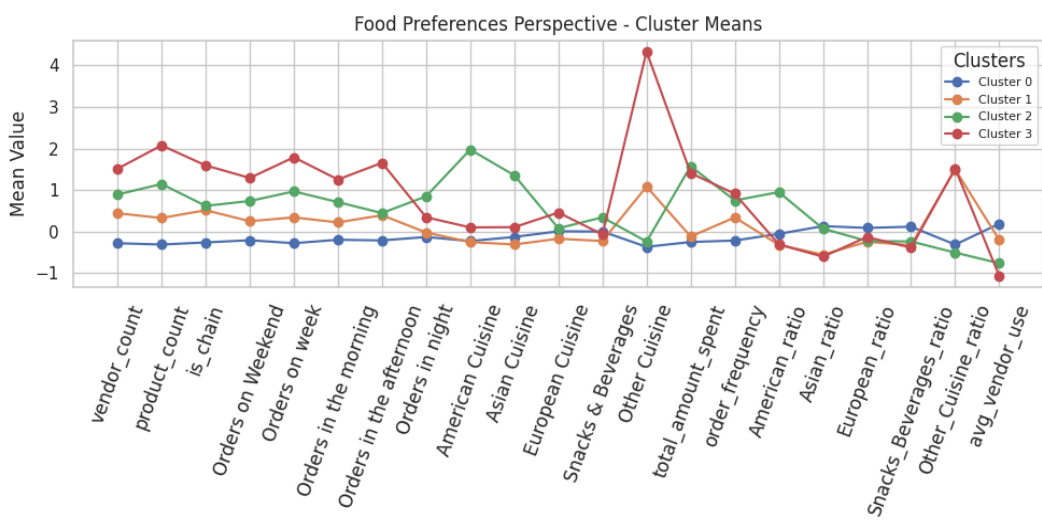
```
df_cluster.groupby('VendorCluster')[Order_features].mean()
```

	vendor_count	product_count	is_chain
VendorCluster			
0	-0.584141	-0.559075	-0.526100
1	0.275513	0.199123	0.182764
2	3.221831	3.432786	3.435518
3	1.457343	1.457023	1.321350

Annex 26 - Final clusters(Behaviour Perspectives)



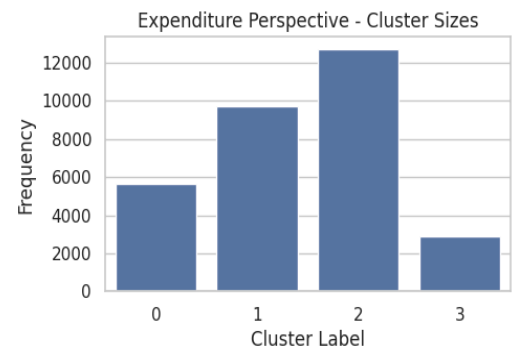
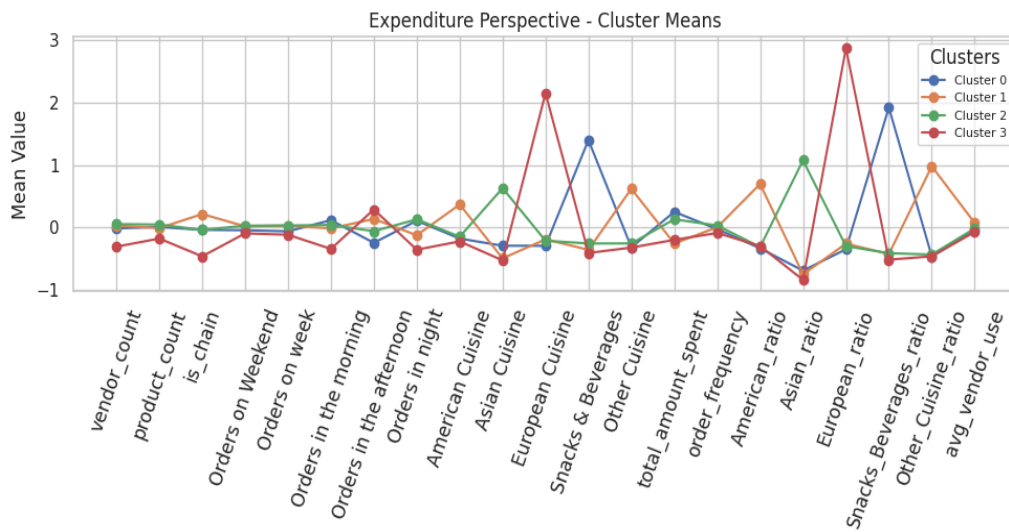
Annex 27 - Final Clusters(Food Perspectives)



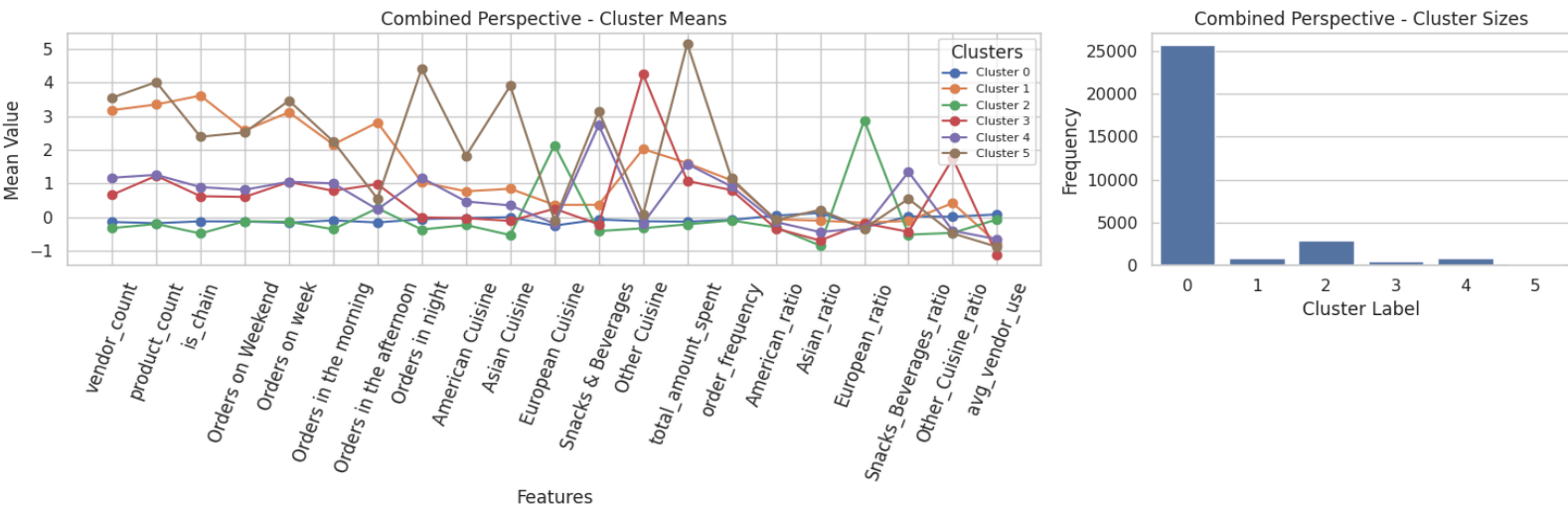
Annex 28 - Final Clusters(Order Perspectives)



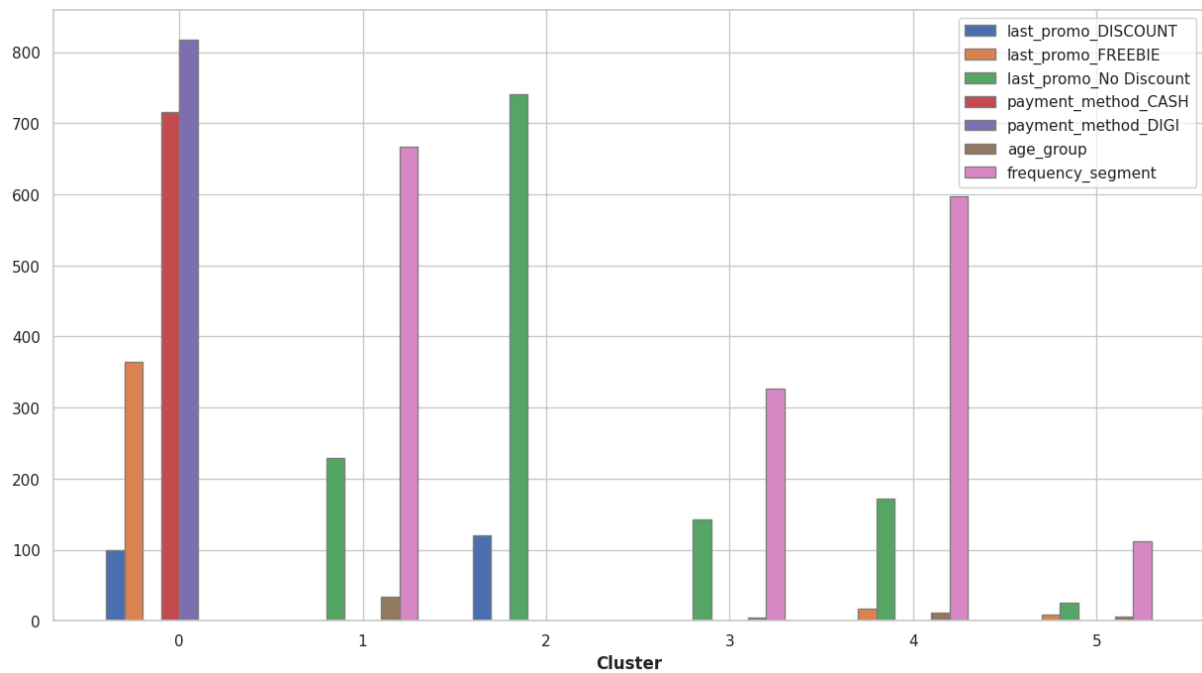
Annex 29 - Final Clusters(Expenditure Perspectives)



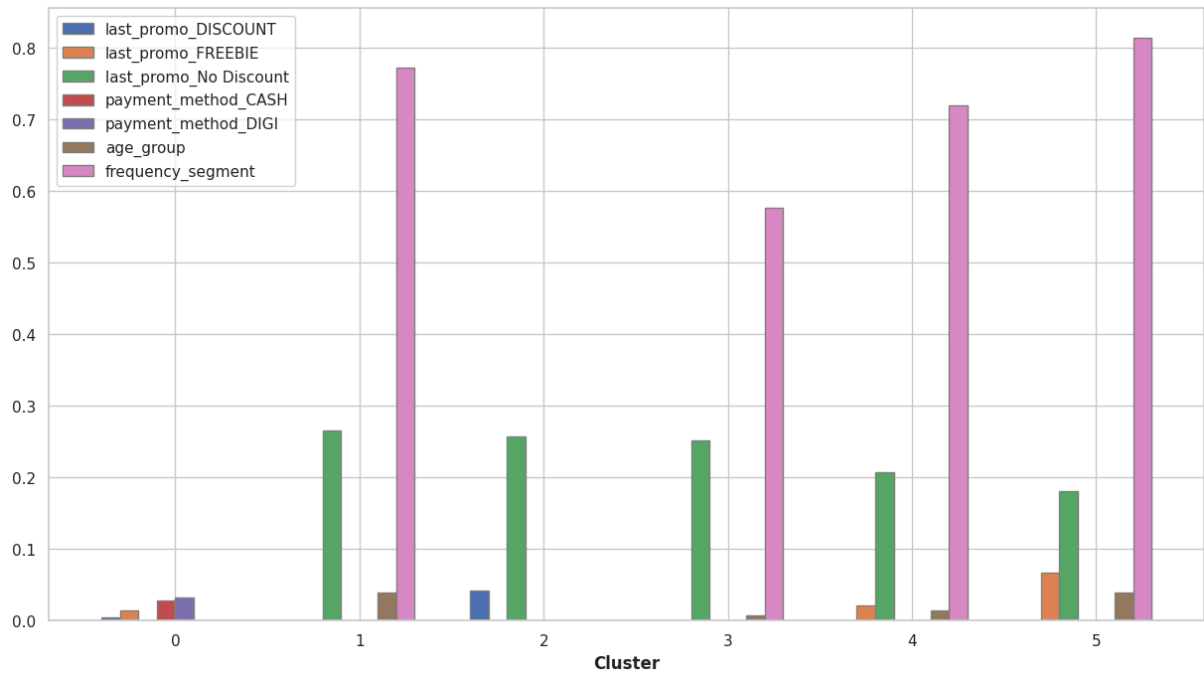
Annex 30 - Final Clusters(Combined Perspectives)



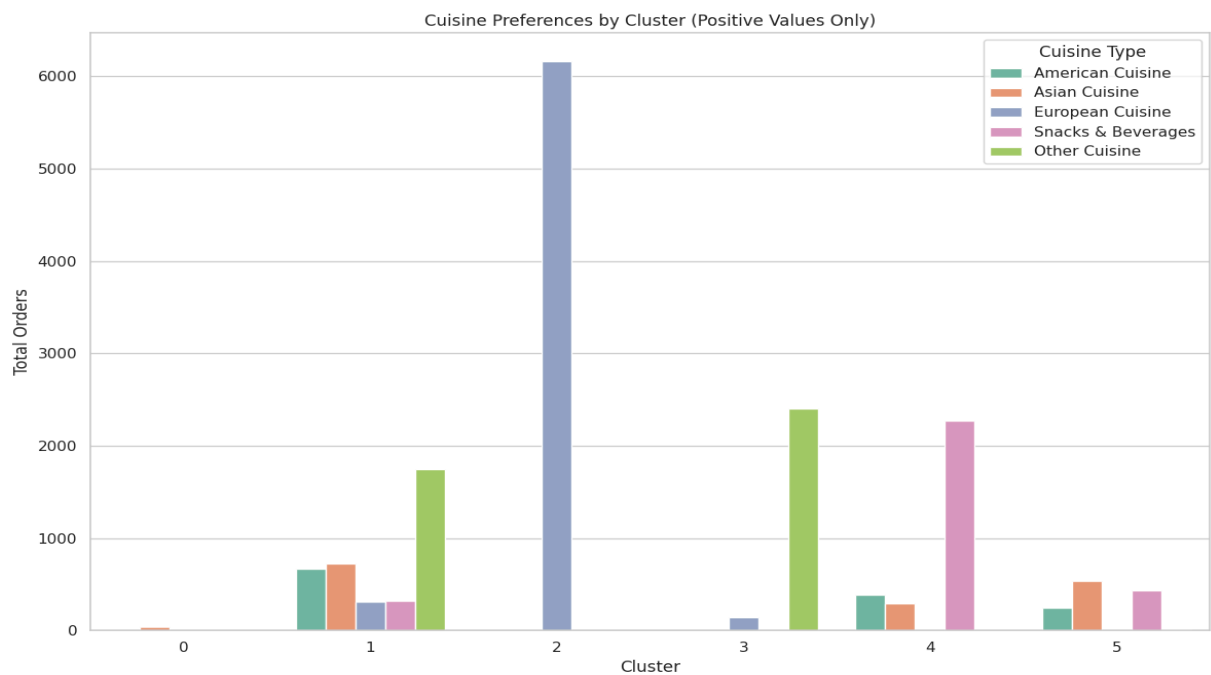
Annex 31 - Categorical Cluster 1



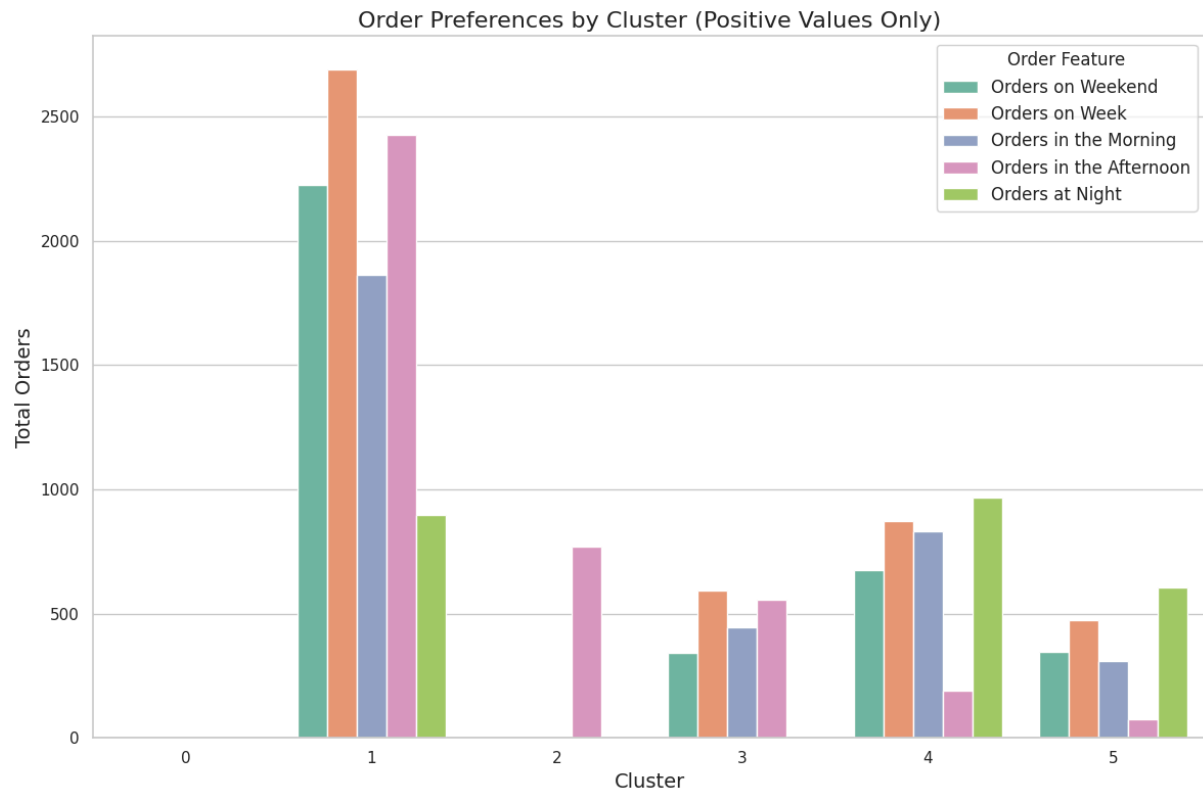
Annex 32 - Categorical Cluster 2



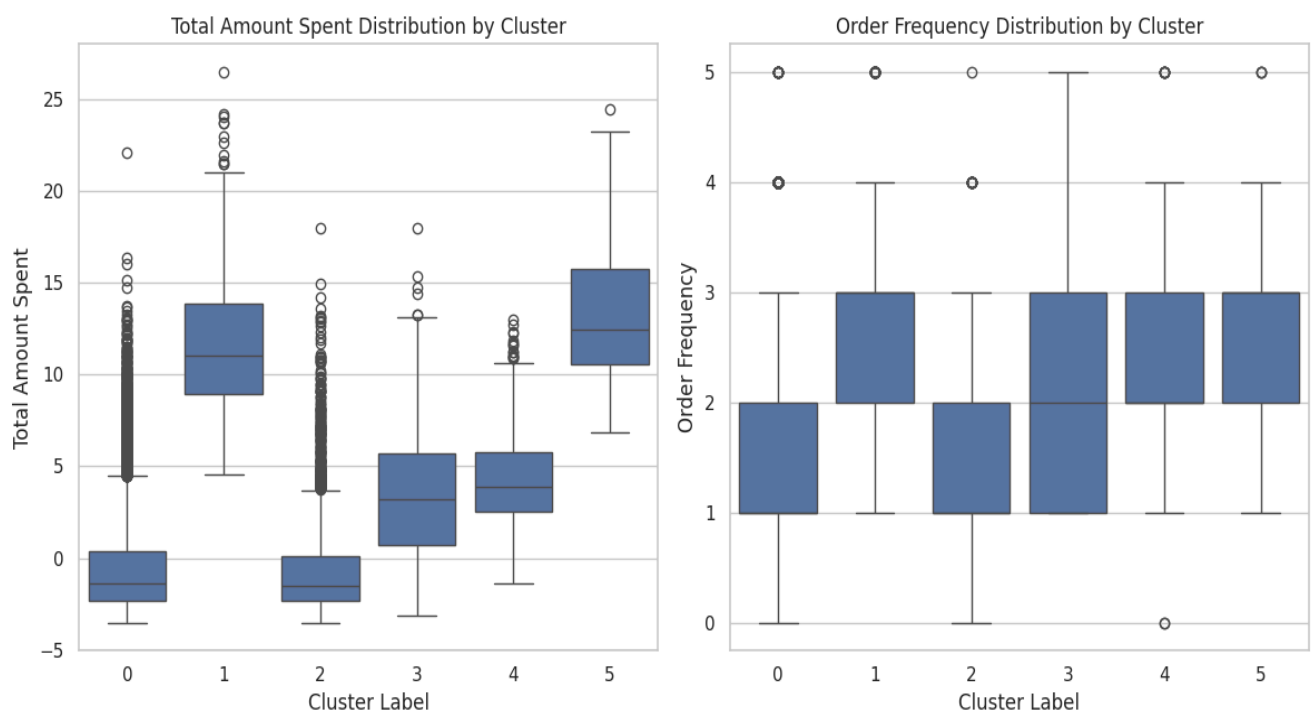
Annex 33 – Order Preferences by Cluster



Annex 34 - Order Preferences by Cluster 2



Annex 35 - Total Amount Spent Distribution by Cluster



Extra Annexes

Annex 15A - Silhouette Scores from the observations(behaviour features)

```
Number of Clusters: 4, Silhouette Score: 0.1947
Number of Clusters: 5, Silhouette Score: 0.1996
Number of Clusters: 6, Silhouette Score: 0.2138
```

Annex 19A - Silhouette Scores from the observations(For food preferences)

```
Number of Clusters: 3, Silhouette Score: 0.2748
Number of Clusters: 4, Silhouette Score: 0.2724
Number of Clusters: 5, Silhouette Score: 0.2279
Number of Clusters: 6, Silhouette Score: 0.2220
```

Annex 19B - R-square score for food features

```
American Cuisine      0.458633
Other Cuisine         0.153605
Asian Cuisine         0.084307
European Cuisine      0.002157
Snacks & Beverages    0.001054
```

Annex 21A - Silhouette Scores from observations(Spending features)

```
Number of Clusters: 3, Silhouette Score: 0.2414
Number of Clusters: 4, Silhouette Score: 0.1854
Number of Clusters: 5, Silhouette Score: 0.1877
Number of Clusters: 6, Silhouette Score: 0.2083
```

Annex 22C - Silhouette Scores from observations(Vendor/Order features)

```
Number of Clusters: 4, Silhouette Score: 0.2153
Number of Clusters: 5, Silhouette Score: 0.2029
Number of Clusters: 6, Silhouette Score: 0.1970
```

Annex 22D - Vendor features R-square contributions

```
is_chain      0.628767
product_count 0.506847
vendor_count  0.476881
dtype: float64
```